



NTNU – Trondheim
Norwegian University of
Science and Technology

TET 4565 - Electricity Markets and Energy System Planning, Specialisation Course

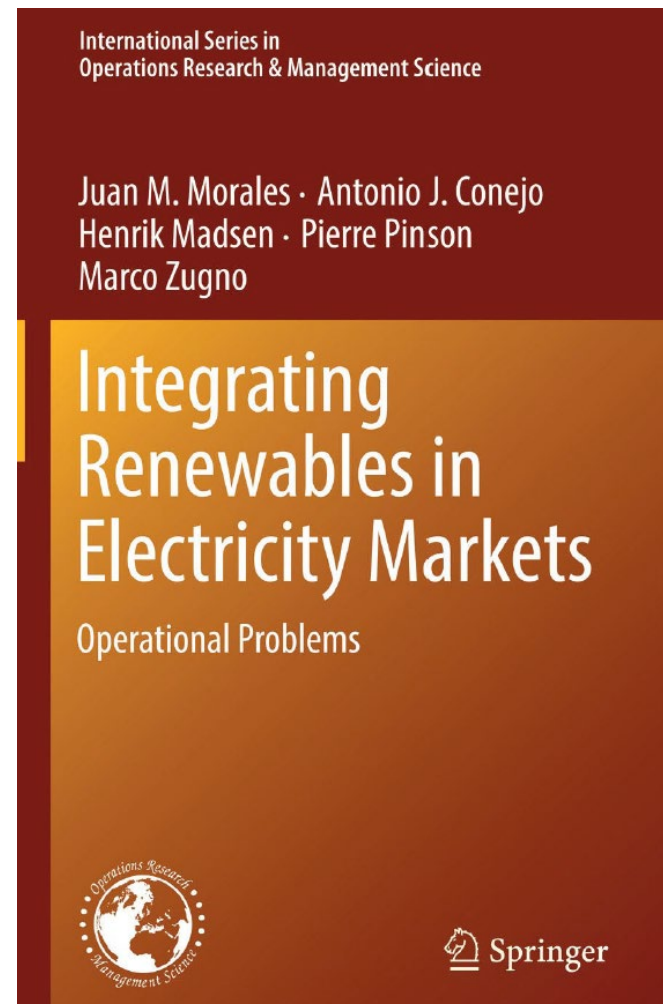
Risk Management for Multi-Market Bidding

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Autumn 2026



Reference

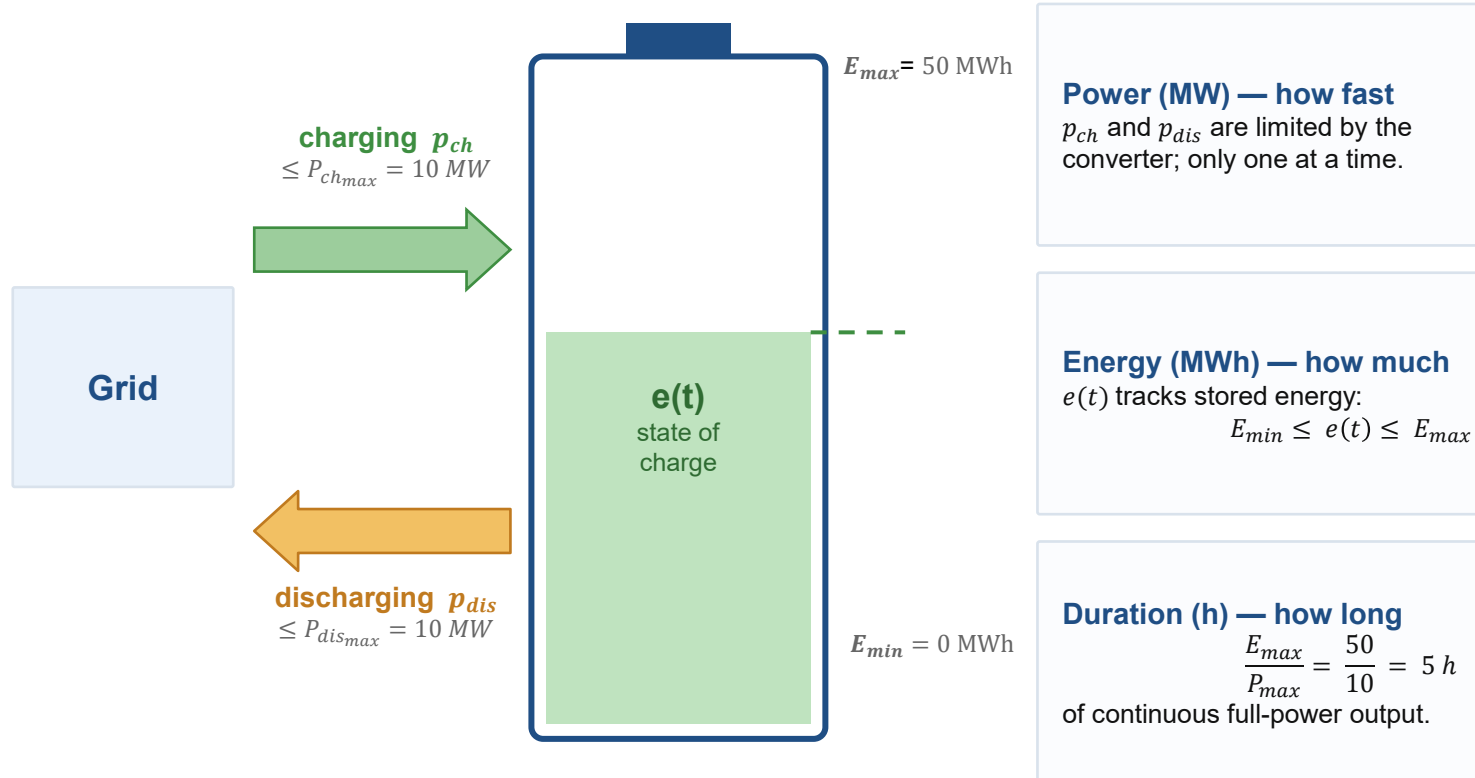
Juan M. Morales, Antonio J. Conejo,
Henrik Madsen, Pierre Pinson, Marco
Zugno, “Integrating Renewables in
Electricity Markets”, 2014, Volume
205, ISBN : 978-1-4614-9410-2



Overview

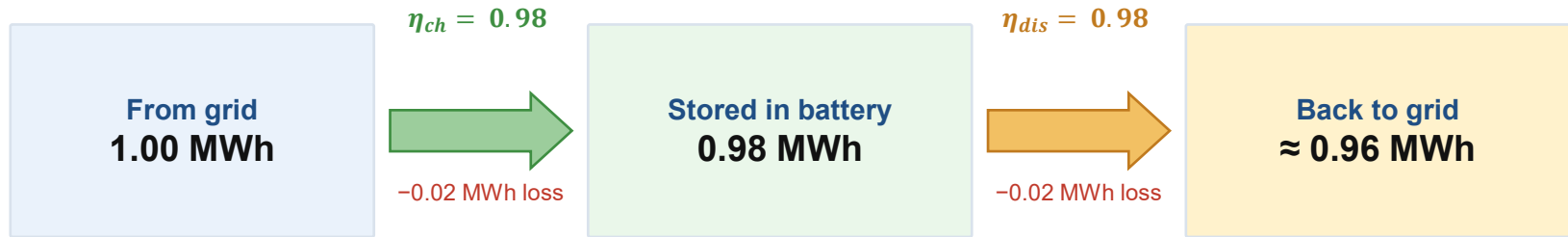
- Battery modelling basics: power, state of charge, efficiency
- Price-taking offering strategy and non-anticipativity
- VaR and CVaR as tail-risk measures
- Battery DA + mFRR multi-market example
- Validation and project-use discussion

Battery basics: power, energy and State of Charge (SOC)



Two ratings define a battery, 10 MW power and 50 MWh energy, and one state variable, $e(t)$, links every decision over time.

Efficiency: every cycle loses some energy



In the SOC equation - charging

1 MWh drawn adds $\eta_{ch} \cdot \Delta \cdot p_{ch} = 0.98$ MWh to the battery.

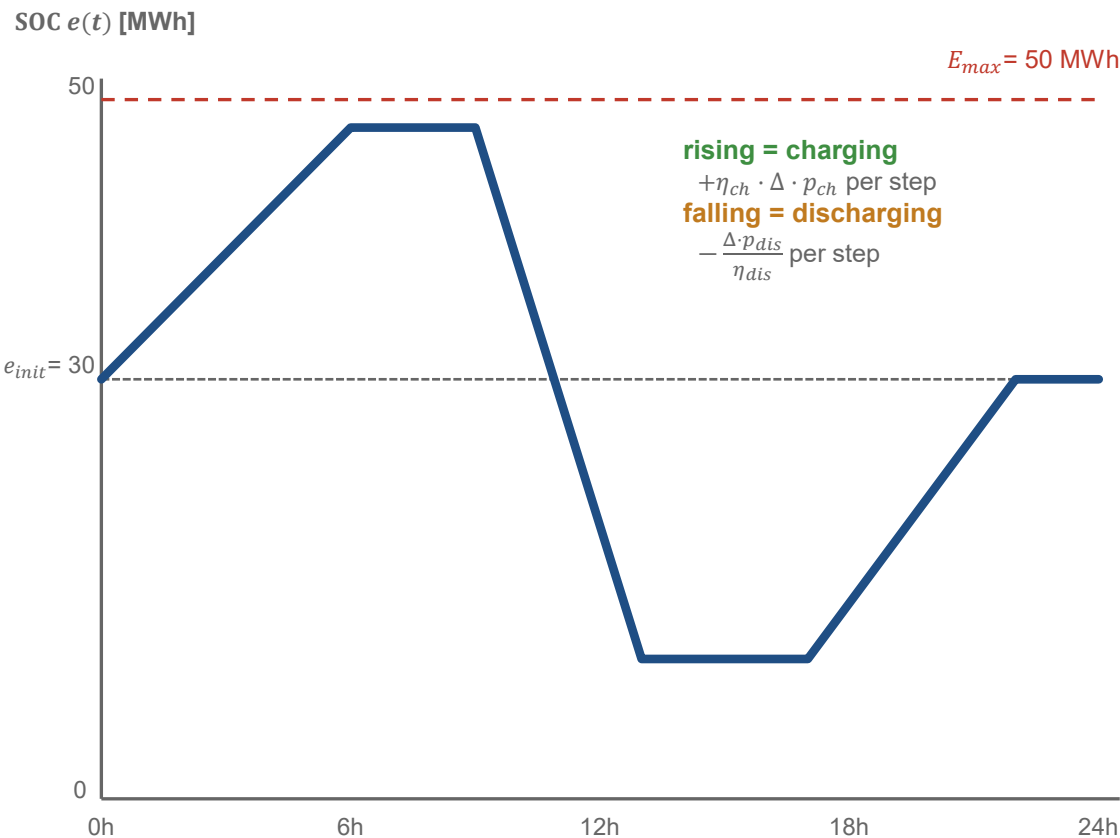
In the SOC equation — discharging

Delivering 1 MWh drains $\frac{\Delta \cdot p_{dis}}{\eta_{dis}} = 1.02$ MWh from the battery.

Round-trip efficiency $\eta_{ch} \cdot \eta_{dis} = 0.98 \times 0.98 \approx 0.96 \rightarrow \approx 4\%$ of every cycle is lost

Consequence for bidding: the price spread earned by a cycle (plus grid tariff) must beat the $\approx 4\%$ round-trip loss.

State of charge: the battery's memory



SOC bookkeeping

$$e[t+1] = e[t] + \eta_{ch} \cdot \Delta \cdot p_{ch} - \frac{\Delta \cdot p_{dis}}{\eta_{dis}}$$

with $\Delta = 0.25$ h and
 $0 \leq e(t) \leq E_{max}$ at all times.

Slope limits: e can change by
 at most $\pm \approx 2.5$ MWh per quarter.

SOC couples all decisions in time and across markets: energy sold now is energy unavailable later.

Frequency-based ancillary services in RGCE and the Nordic



Abbreviation	Full name
FFR	Fast Frequency Reserve
FCR-N	Frequency Containment Reserve – Normal
FCR-D	Frequency Containment Reserve – Disturbance
aFRR	automatic Frequency Restoration Reserve
mFRR	manual Frequency Restoration Reserve

Nordic battery reserve products

Activation time, endurance, direction, minimum bid and main battery constraints - Norway

Product	Activation time	Endurance	Direction	Min. bid	Main battery constraint
FFR (Profile)	0.7, 1.0 or 1.3 s	5 or 30 s	Up	1 MW	Fixed seasonal availability hours
FFR (flex)	0.7, 1.0 or 1.3 s	5 or 30 s	Up	5 MW	Weekly call-off / uncertain availability requirement
FCR-D	86% within 7.5 s	≥20 min	Up/down (separate)	0.1 MW	Energy endurance + counter-direction power reservation
FCR-N	63% within 60 s; 95% within 3 min	≥1 h in both directions	Symmetric	0.1 MW	SoC management, energy availability and frequent activation
aFRR	Within 5 min	1 h	Up/down (separate)	1 MW	Sustained activation, communication and energy availability
mFRR	12.5 min full activation	15 min scheduled / up to 30 min direct activation	Up/down (separate)	1 MW	Headroom/footroom, SoC and energy deliverability

Source: Nora Ingvild Gulbrandsen, "Improving Battery Storage Profitability through Value Stacking in Norway's Day-Ahead and mFRR Markets", Master thesis, 2026

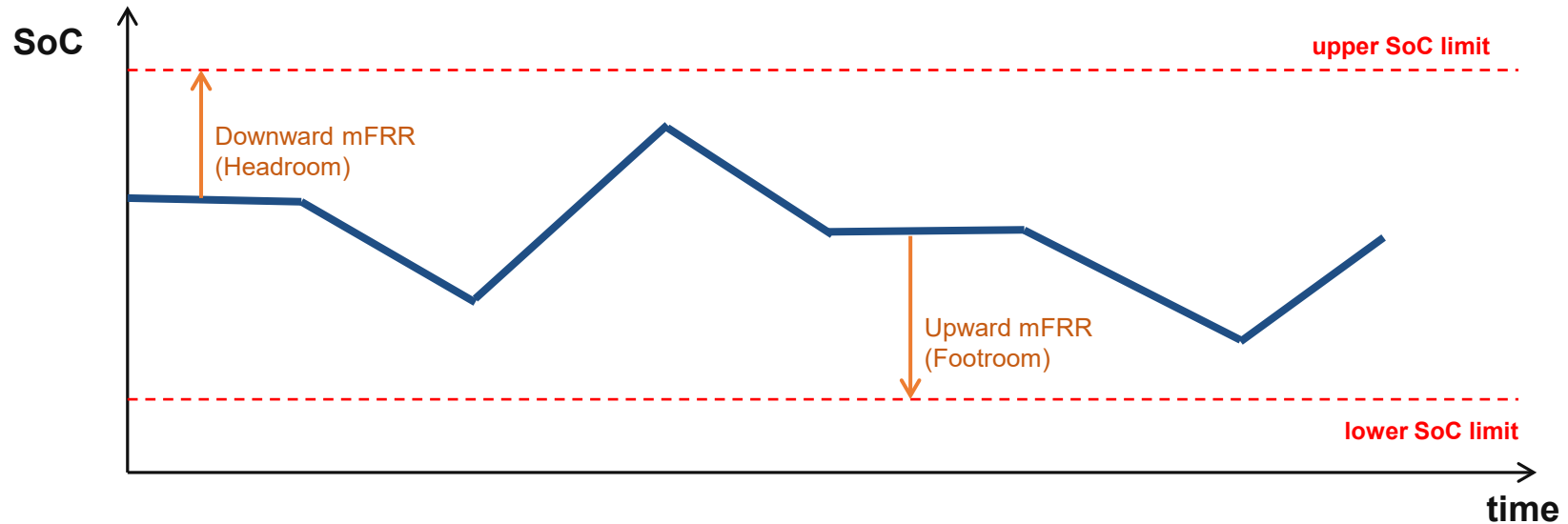
Battery bidding is constrained by state of charge

10 MW / 50 MWh example

5-hour nominal duration

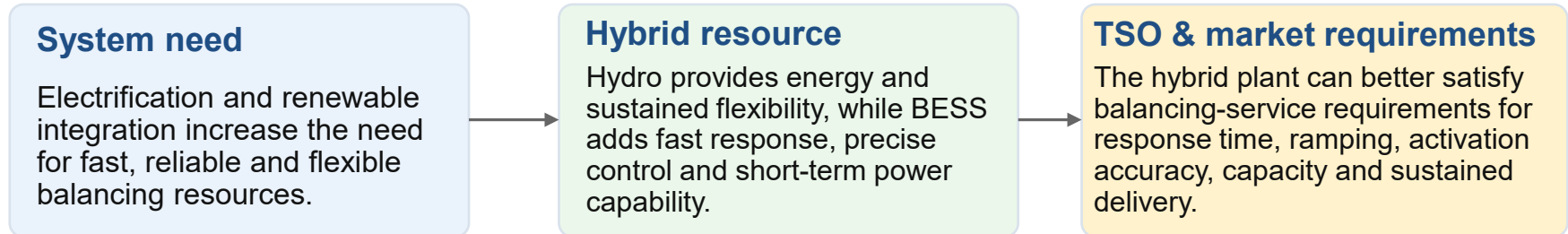
Reserve deliverability

Upward mFRR needs energy; downward mFRR needs free capacity



Value stacking is constrained: energy sold in one market cannot be freely sold again in another.

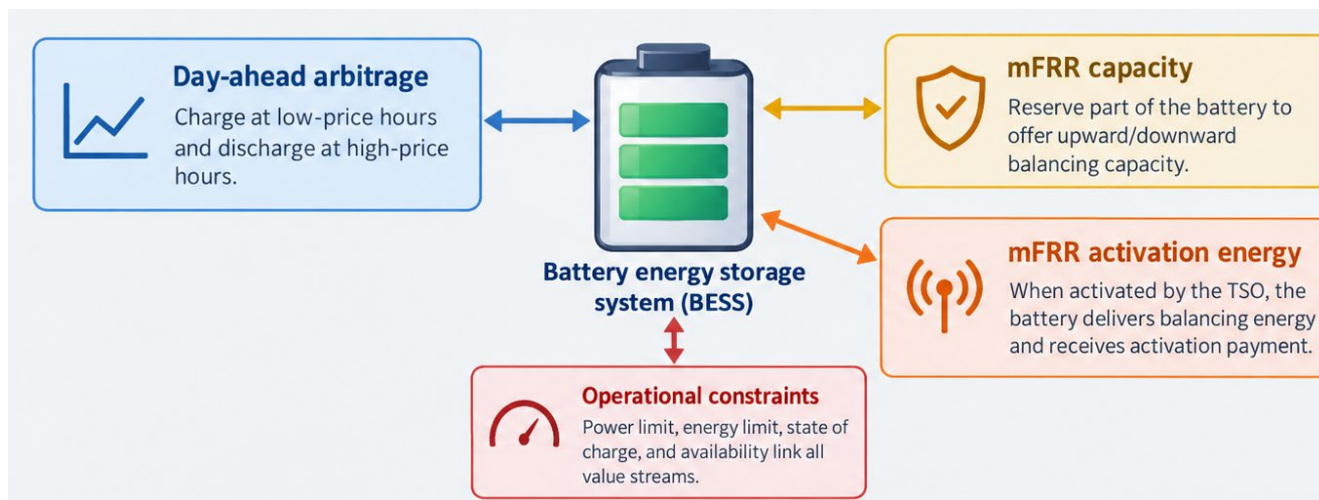
BESS (battery energy storage system)-Hybrid sources



Why hybridization matters:

- The battery helps the hydropower plant deliver rapid activation and track TSO signals more accurately.
- Hydropower provides energy and endurance, while the battery handles fast transients and short-duration imbalances.
- Co-optimizing water release, battery power, and state of charge improves reserve availability and operational feasibility.
- Uncertain activation creates both revenue risk and feasibility risk, making coordinated scheduling essential.

BESS value stacking — Nordic day-ahead + mFRR



Conceptual logics

- The same battery can participate in both the day-ahead market and balancing services
- Day-ahead scheduling determines charging and discharging opportunities
- mFRR capacity bids create availability obligations and may limit arbitrage freedom in the day-ahead market.
- If activated, mFRR energy delivery changes SoC and affects later market opportunities

Illustrative stacked value

DA arbitrage

mFRR capacity payment

mFRR activation energy

Total battery value = energy arbitrage + reserve capacity payment + activation payment

Bidding-zone dependence: same model, different value



Multi-market bidding is not one universal answer. The optimal bid depends on local price distributions, activation tails, and SoC constraints.

NO3

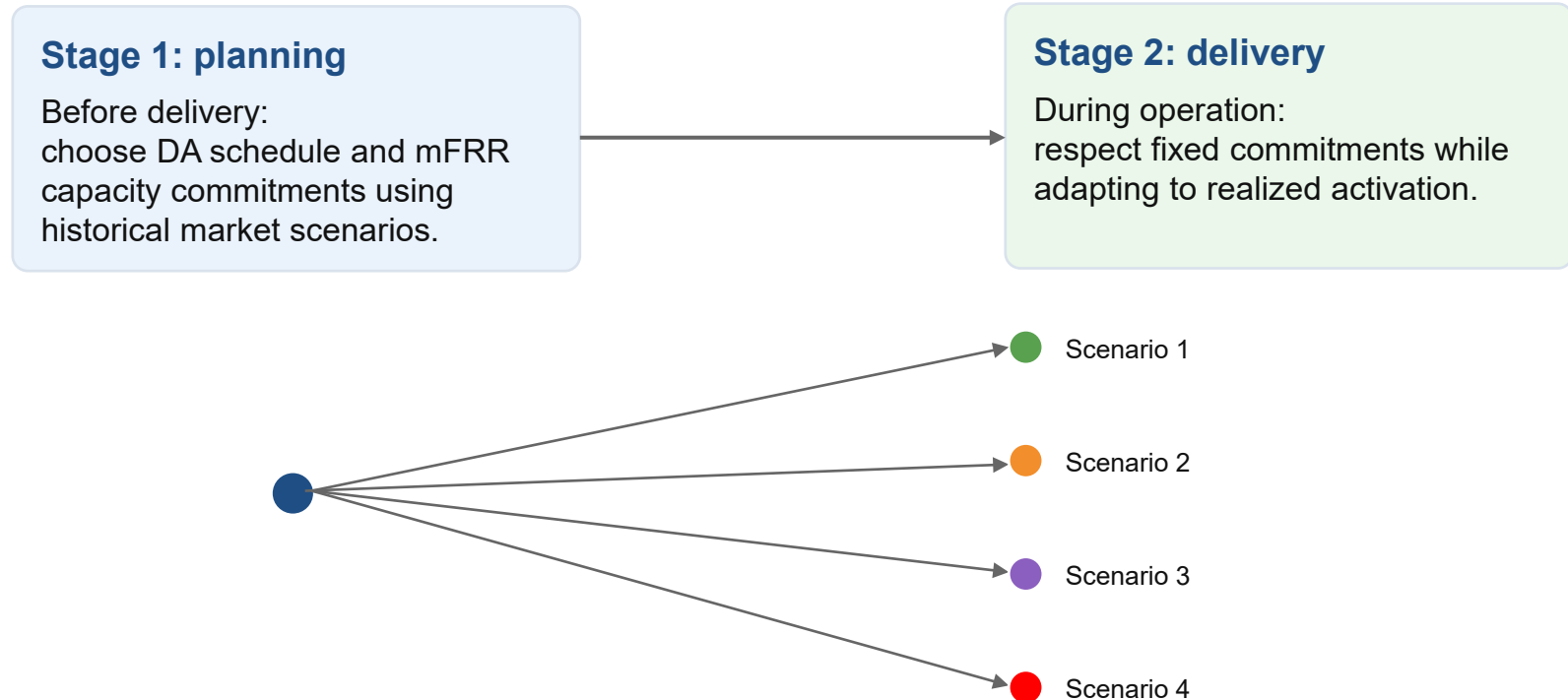
Lower DA price level and lower median mFRR activation prices, but important upward price spikes.

NO2

Higher price level and stronger value-stacking.

This is why scenario construction and out-of-sample testing matter.

Applied framework: two-stage planning and delivery



First-stage decisions are not scenario-indexed; second-stage recourse adapts to realized conditions.

Source: Nora Ingvild Gulbransen, "Improving Battery Storage Profitability through Value Stacking in Norway's Day-Ahead and mFRR Markets", Master thesis, 2026

Value stacking is not automatically profitable

Bidding Zone	DA only	DA + mFRR capacity	Full DA + mFRR EAM
NO3	22.2 kEUR	16.6 kEUR	78.9 kEUR
NO2	305.9 kEUR	—	973.0 kEUR

Opportunity cost of reserves

In NO3, adding mFRR capacity alone reduced net profit because reserve commitments limited DA arbitrage.

Activation value

When voluntary mFRR activation was included, net profit increased strongly.

Risk-aware bidding asks: how much expected profit should we sacrifice to avoid bad tail outcomes and infeasible activation?

Source: Nora Ingvild Gulbrandsen, "Improving Battery Storage Profitability through Value Stacking in Norway's Day-Ahead and mFRR Markets", Master thesis, 2026

Where this lecture fits



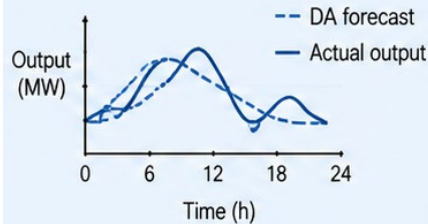
Keep the two-stage scenario model, but replace "best in expectation" by "profitable enough in the bad tail".

Risk management asks: are we comfortable with the worst outcomes of the selected bid?

Why risk is unavoidable in renewable bidding

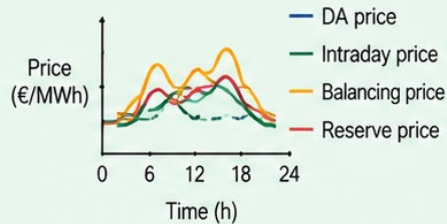
Volume risk

Wind/solar output differs from the DA forecast.



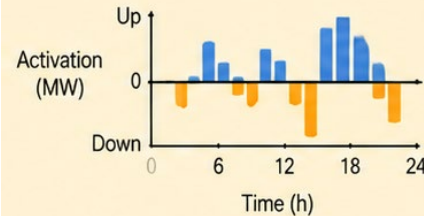
Price risk

DA, reserve, intraday and balancing prices are uncertain.



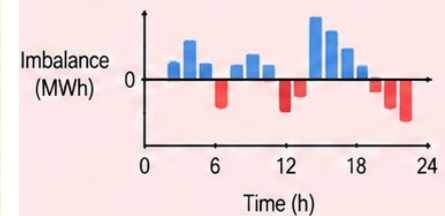
Activation risk

Reserve capacity can be activated with uncertain direction and volume.



Penalty risk

Imbalances can trigger costly settlement.



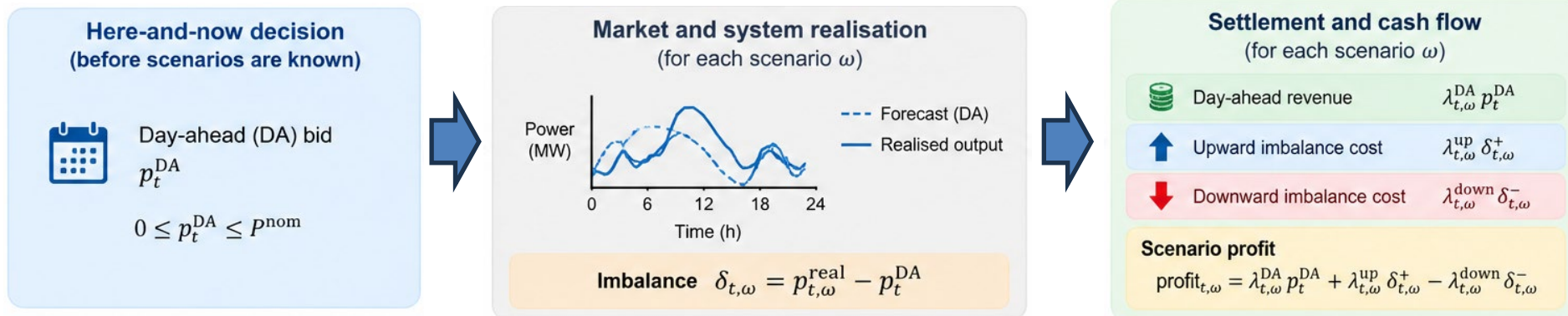
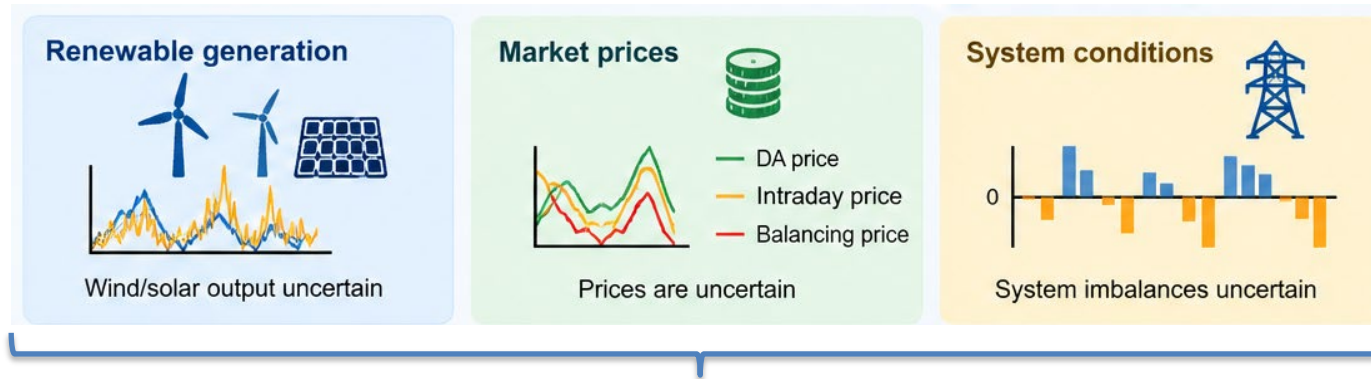
All sources are translated into scenario cash flow



Profit $[\omega]$ = market revenues $[\omega]$ - imbalance costs $[\omega]$ - penalties $[\omega]$

Optimizing in an uncertain market environment

Source of uncertainty (scenario $\omega \in \Omega$)



Objective: Maximise expected profit (risk-neutral)

$$\max_{p^{DA}} \sum_{\omega \in \Omega} \pi_{\omega} \text{profit}_{\omega}$$

Key features

- Here-and-now DA decision
- Scenario-dependent imbalances
- Risk-neutral: only probability-weighted mean matters

Generating price scenarios

Good scenarios should reproduce price levels, volatility, spikes, temporal dependence and cross-market correlation.

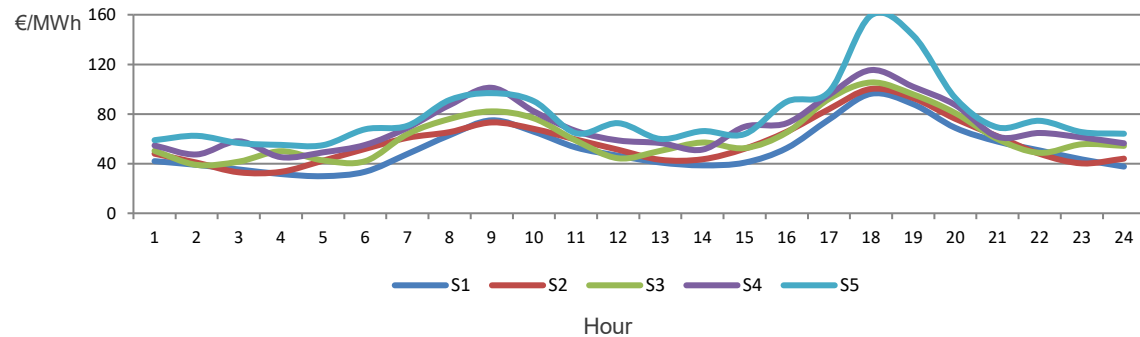
Historical sampling

Reuse past days or sample similar days by season, demand, wind/hydro and market conditions.

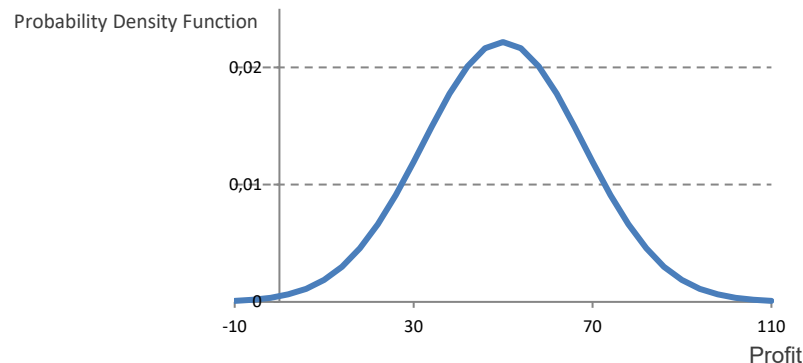
Time-series simulation

Fit AR/VAR/GARCH, residual or quantile models and simulate many hourly price paths.

Historic / sampled DA price scenarios



Normal approximation of scenario profit



Nomenclature: risk-neutral DA + balancing model

Classical OR notation: sets, parameters, variables

Sets and indices	
Symbol	Description
$t \in \mathcal{T}$	Time periods, e.g. hours or market time units
$\omega \in \Omega$	Uncertainty scenarios for production, prices and activation

Decision and auxiliary variables	
Variable	Meaning / domain
p_t^{DA}	Day-ahead bid; here-and-now decision, $0 \leq p_t^{DA} \leq p^{nom}$ [MW]
$\delta_{t,\omega}$	Net imbalance after uncertainty is revealed [MW]
$\delta_{t,\omega}^+, \delta_{t,\omega}^-$	Positive and negative imbalance parts, both ≥ 0 [MW]
$profit_{t,\omega}$	Profit contribution in period t and scenario ω [€]

Parameters		
Parameter	Description	Unit
π_ω	Probability weight of scenario ω	-
$p_{t,\omega}^{real}$	Realized renewable production in scenario ω	MW
p^{max}	Installed / nominal capacity limit	MW
$\lambda_{t,\omega}^{DA}$	Day-ahead electricity price	€/MWh
$\lambda_{t,\omega}^{up}$	Upward-imbalance settlement price	€/MWh
$\lambda_{t,\omega}^{down}$	Downward-imbalance settlement price	€/MWh

Note: realised production and prices are treated as scenario-dependent parameters, not decision variables.

Risk-neutral benchmark: DA + balancing

Maximize expected profit:

$$\max_{p_t^{\text{DA}}, \delta_{t,\omega}^+, \delta_{t,\omega}^-} \sum_{\omega \in \Omega} \pi_{\omega} \text{profit}_{t,\omega}$$

Scenario profit:

$$\text{profit}_{t,\omega} = \lambda_{t,\omega}^{\text{DA}} p_t^{\text{DA}} + \lambda_{t,\omega}^{\text{up}} \delta_{t,\omega}^+ - \lambda_{t,\omega}^{\text{down}} \delta_{t,\omega}^- \quad \forall t \in \mathcal{T}, \omega \in \Omega,$$

Constraints:

$$\begin{aligned} 0 &\leq p_t^{\text{DA}} \leq p^{\text{max}}, \quad \forall t \in \mathcal{T}, \\ \delta_{t,\omega} &= p_{t,\omega}^{\text{real}} - p_t^{\text{DA}}, \quad \forall t \in \mathcal{T}, \omega \in \Omega, \\ \delta_{t,\omega} &= \delta_{t,\omega}^+ - \delta_{t,\omega}^- \quad \forall t \in \mathcal{T}, \omega \in \Omega \\ \delta_{t,\omega}^+ &\geq 0, \quad \delta_{t,\omega}^- \geq 0, \quad \forall t \in \mathcal{T}, \omega \in \Omega \end{aligned}$$

Here-and-now

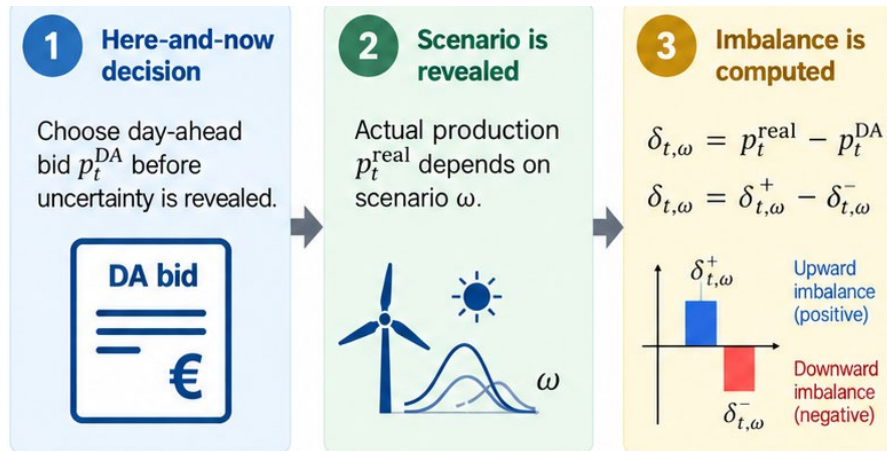
p_t^{DA} is chosen before the scenario is revealed (**Non-anticipativity**).

Wait-and-see

Imbalance variables are scenario dependent.

Risk-neutral

Only the probability-weighted mean matters.

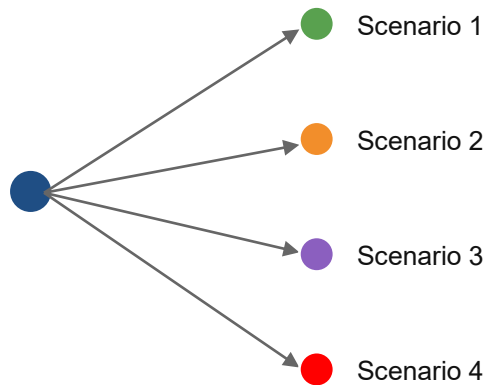


Here, $\lambda_{t,\omega}^{\text{DA}}$, $\lambda_{t,\omega}^{\text{up}}$, and $\lambda_{t,\omega}^{\text{down}}$ represent the day-ahead, upward-imbalance, and downward-imbalance prices, respectively.

One bid - many possible real-time outcomes

Day-ahead stage

Balancing stage



Non-anticipativity:

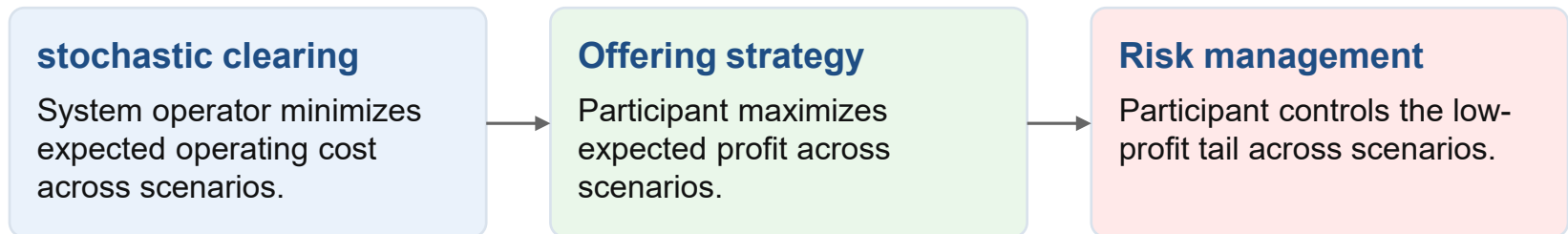
p_t^{DA} is the same
for every scenario w .

Recourse:

$\delta_{t,\omega}^+$ and $\delta_{t,\omega}^-$ adapt after uncertainty is
revealed.

Risk-aware bidding changes the first-stage bid by considering the tail of scenario profits.

Same two-stage structure - new economic question



Risk-neutral objective:

$$\max_x \mathbb{E}[\text{profit}(x)]$$

Optimizes the probability-weighted mean

Risk-aware objective:

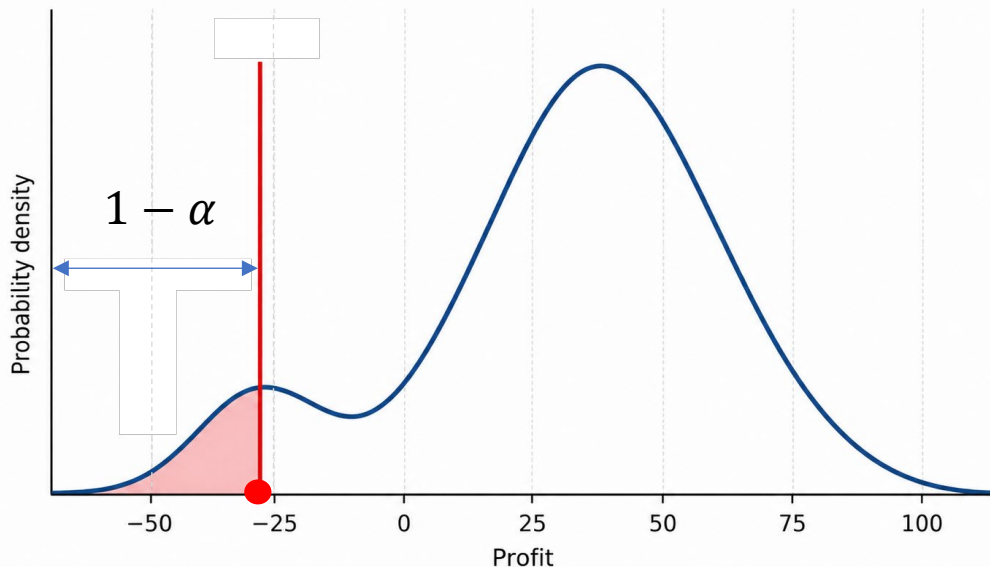
$$\max_x \mathbb{E}[\text{profit}(x)] + \text{tail protection}$$

Improves robustness against bad scenarios.

Expected profit alone can hide scenarios that threaten solvency, contracts, or project economics.

Value at Risk (VaR): where the downside tail begins

The Probability Density Function (PDF) of the Battery's profit might look as follows:



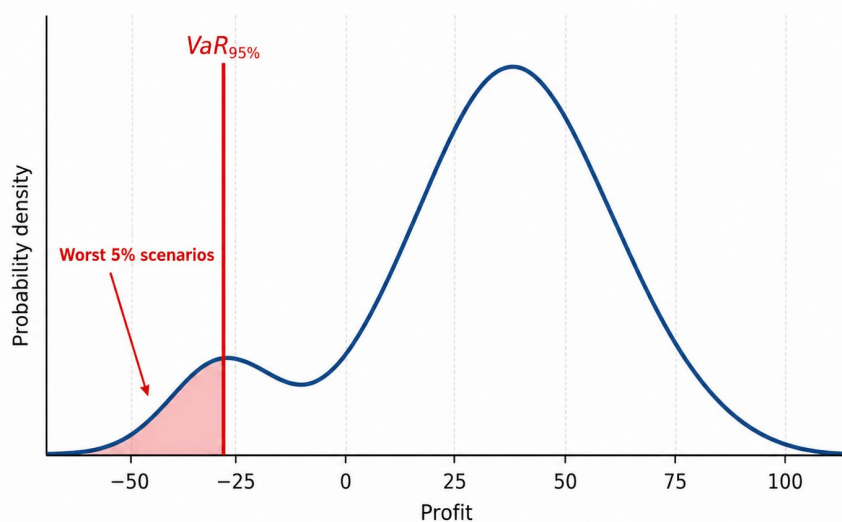
VaR identifies the threshold separating the worst $1-\alpha$ fraction of profit outcomes from the rest of the distribution.

$(1 - \alpha)$ -quantile of the distribution, where $0 \leq \alpha \leq 1$

VaR (Value-at-Risk) is the minimum profit such that the probability of the profit being lower than this value is less than or equal to $1-\alpha$.

Value at Risk (VaR): where the downside tail begins

VaR is a profit threshold: only a small probability of scenarios falls below it.



Definition

$$\text{VaR}_{95\%}(\pi) = q_{5\%}(\pi)$$

$$\Pr(\pi \leq \text{VaR}_{\alpha}) \approx 1 - \alpha$$

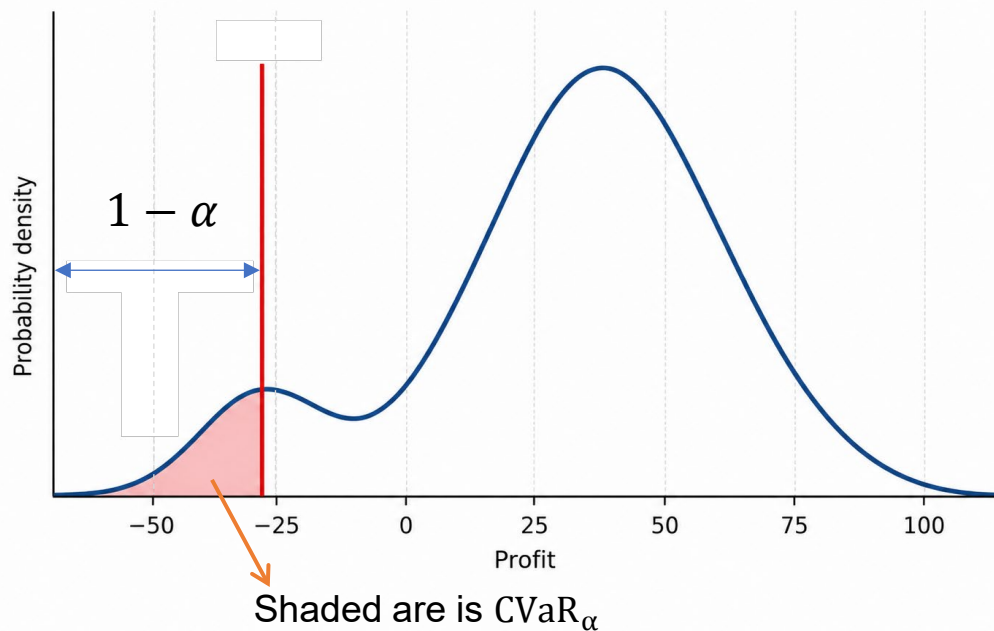
Interpretation for profit

For $\text{VaR}_{95\%} \approx -28$, about 5% of scenarios result in a profit below -28

VaR identifies where the bad tail begins, but it does not measure how severe the outcomes beyond that threshold are.

Conditional value-at-risk (CVaR)

The Probability Density Function (PDF) of the Battery's profit might look as follows:



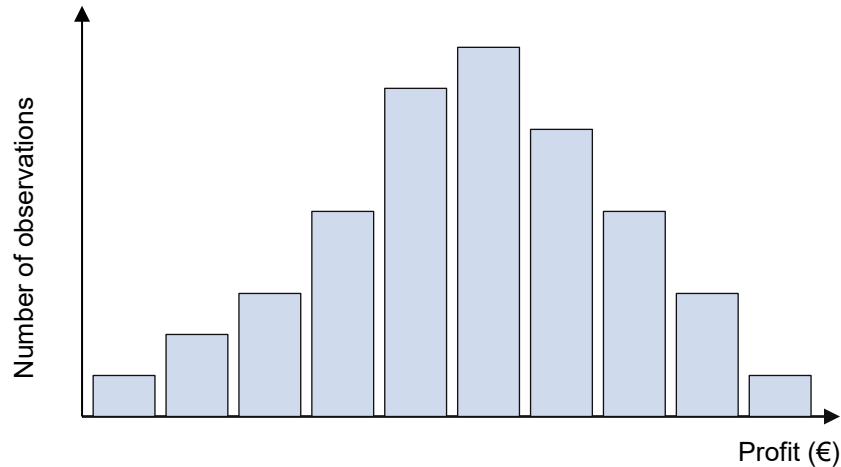
CVaR (Conditional Value-at-Risk) is the average profit in the worst $1-\alpha$ fraction of scenarios

In other words, CVaR is the expected profit in the tail region where profit is below the VaR threshold.

Unlike VaR, CVaR is a key risk measure when the underlying distribution has a 'fat tail' (i.e., a larger tail region, or gray area, where extreme losses occur).

Conditional value-at-risk (CVaR):

Let's now consider a discrete distribution for multi-market profit:



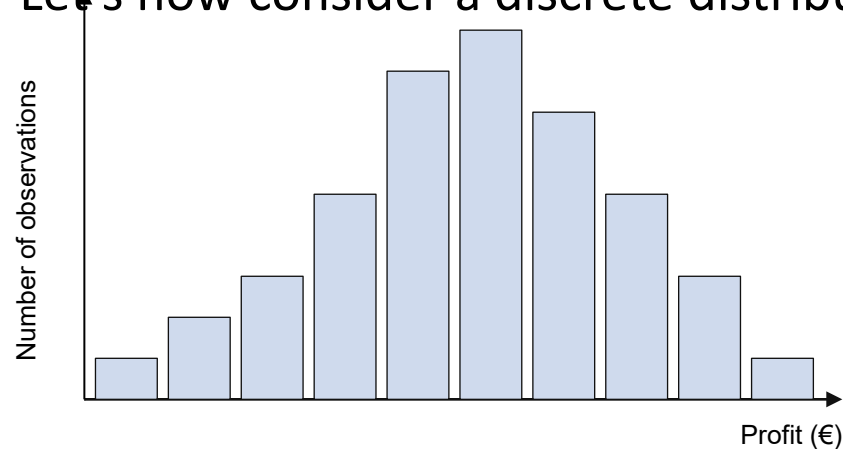
Maximization of the **expected profit**:

$$\underset{P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}}{\text{maximize}} \quad \sum_{\omega} \pi_{\omega} \text{Profit}_{\omega}(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow})$$

$$\begin{aligned} \text{subject to} \quad & f(P^{\text{DA}}) \leq 0, \\ & g(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \leq 0, \quad \forall \omega. \end{aligned}$$

Conditional value-at-risk (CVaR):

Let's now consider a discrete distribution for multi-market profit:



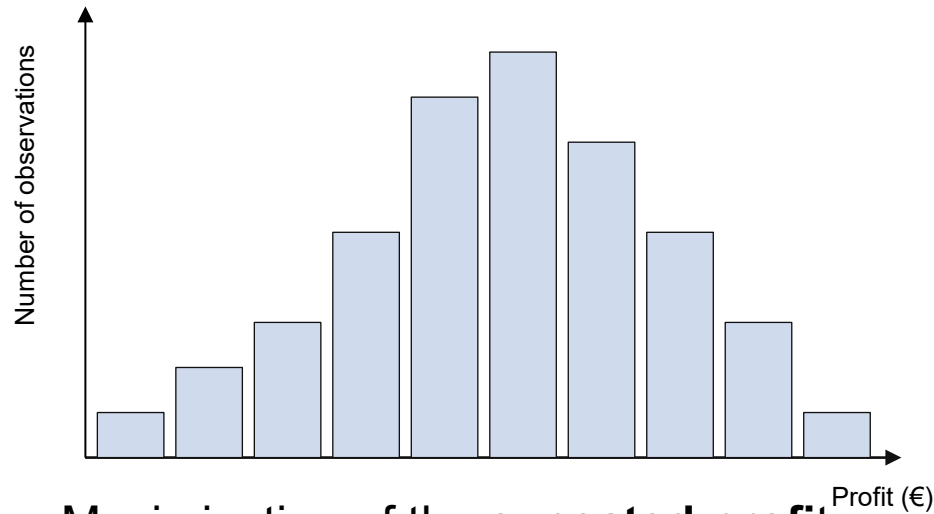
Maximization of the **expected profit**:

$$\begin{aligned}
 &\underset{P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}}{\text{maximize}} && \sum_{\omega} \pi_{\omega} \text{Profit}_{\omega}(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \\
 & && \text{Day-ahead offer (MW)} \quad \text{offer (MW)} \quad \text{Imbalances under scenario } \omega \text{ (MW)} \\
 &\text{subject to} && f(P^{\text{DA}}) \leq 0, \\
 & && g(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \leq 0, \quad \forall \omega.
 \end{aligned}$$

Technical constraints

Conditional value-at-risk (CVaR):

Let's now consider a discrete distribution for multi-market profit:

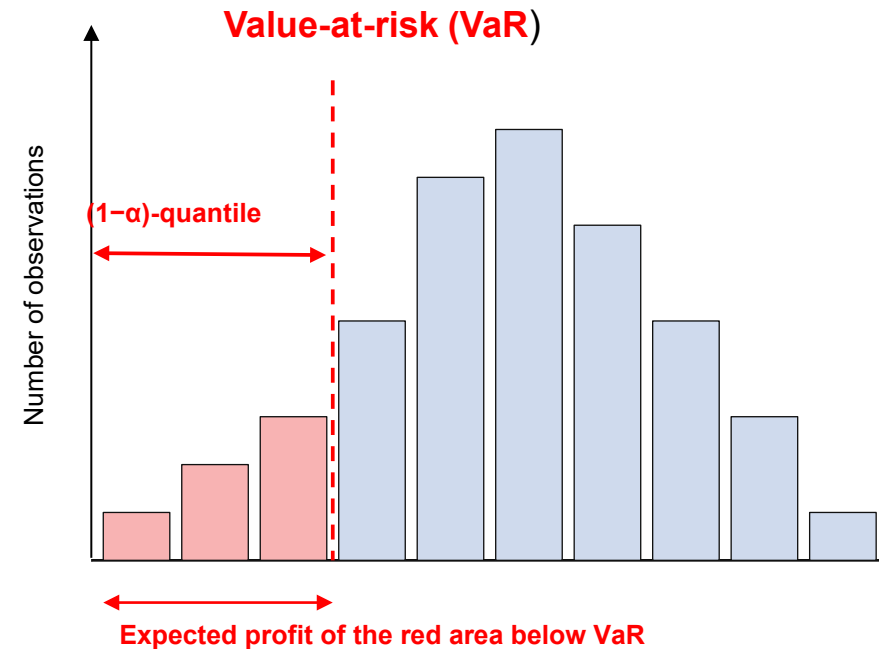


Maximization of the **expected profit**:

$$\text{maximize}_{P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}} \sum_{\omega} \underbrace{\pi_{\omega}}_{\text{Probability of scenario } \omega} \underbrace{\text{Profit}_{\omega}(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow})}_{\text{Profit function under scenario } \omega \text{ (€)}}$$

$$\text{subject to } f(P^{\text{DA}}) \leq 0, \\ g(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \leq 0, \quad \forall \omega.$$

Conditional value-at-risk (CVaR): average of the bad tail

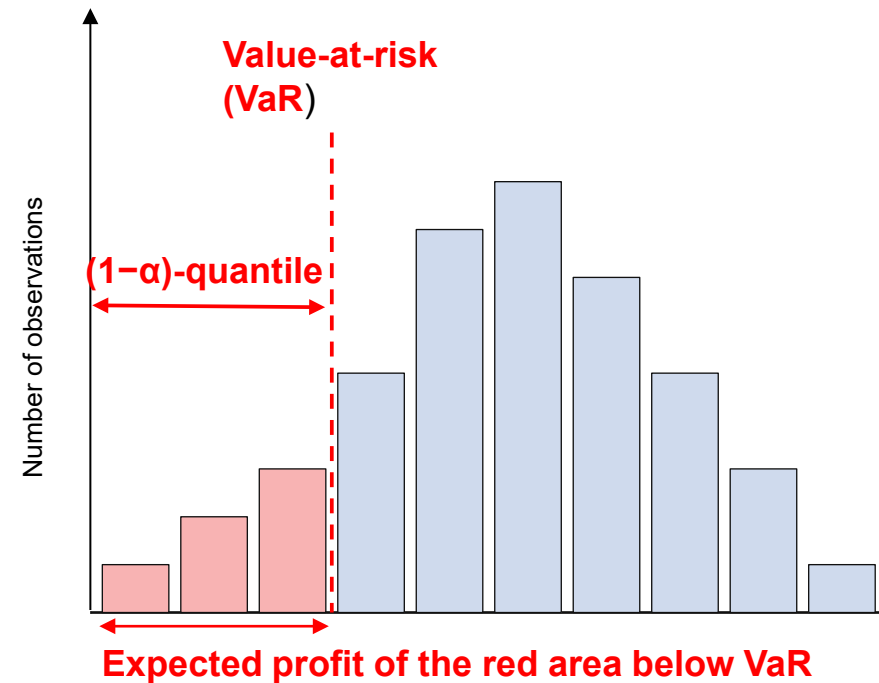


Maximization of the **CVaR**::

$$\begin{aligned}
 & \underset{P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}, \zeta}{\text{maximize}} && \underbrace{\zeta - \frac{1}{1-\alpha} \sum_{\omega} \pi_{\omega} \max[0, \zeta - \text{Profit}_{\omega}(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow})]}_{\text{The definition of CVaR (€) according to Rockafellar and Uryasev [1]}} \\
 & \text{subject to} && f(P^{\text{DA}}) \leq 0, \\
 & && g(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \leq 0, \quad \forall \omega.
 \end{aligned}$$

[1] Rockafellar, R.T., Uryasev, S., "Optimization of conditional value-at-risk," *Journal of Risk*, vol. 2, pp. 21–41, 2000

Conditional value-at-risk (CVaR): average of the bad tail



Maximization of the **CVaR**::

$$\text{maximize}_{P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}, \zeta} \quad \zeta - \frac{1}{1-\alpha} \sum_{\omega} \pi_{\omega} \max[0, \zeta - \text{Profit}_{\omega}(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow})]$$

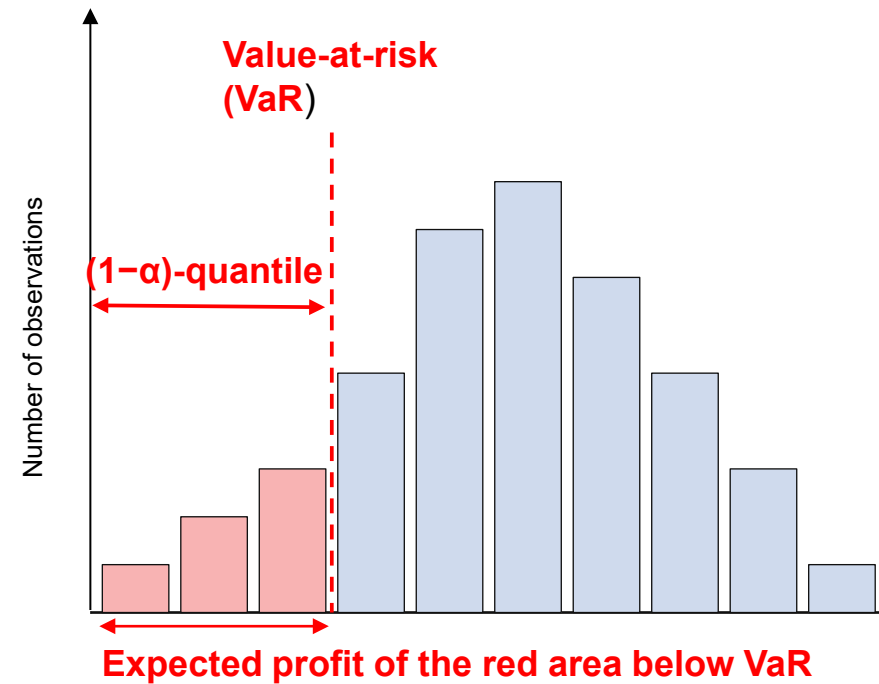
$$\text{subject to} \quad f(P^{\text{DA}}) \leq 0,$$

$$g(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \leq 0, \quad \forall \omega.$$

VaR (€) (it is a variable to be determined by the optimization problem)

Note that the optimal value of VaR is determined endogenously (it is not a parameter).

Conditional value-at-risk (CVaR): average of the bad tail

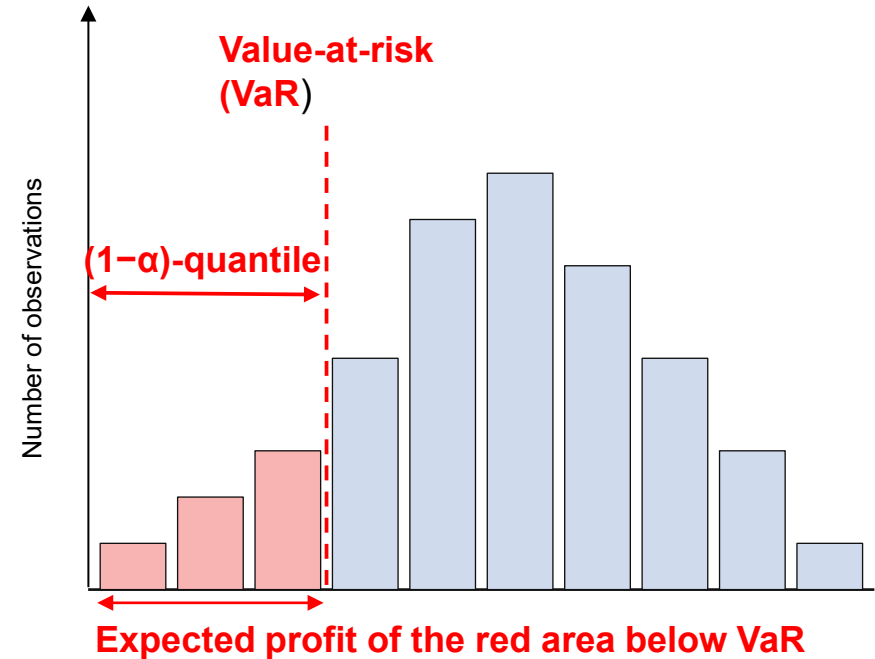


Maximization of the **CVaR**::

$$\begin{aligned}
 & \underset{P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}, \zeta}{\text{maximize}} && \zeta - \frac{1}{1-\alpha} \sum_{\omega} \pi_{\omega} \max[0, \zeta - \text{Profit}_{\omega}(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow})] \\
 & \text{subject to} && f(P^{\text{DA}}) \leq 0, \\
 & && g(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \leq 0, \quad \forall \omega.
 \end{aligned}$$

How to reformulate the term **max[.,.]** in the objective function?

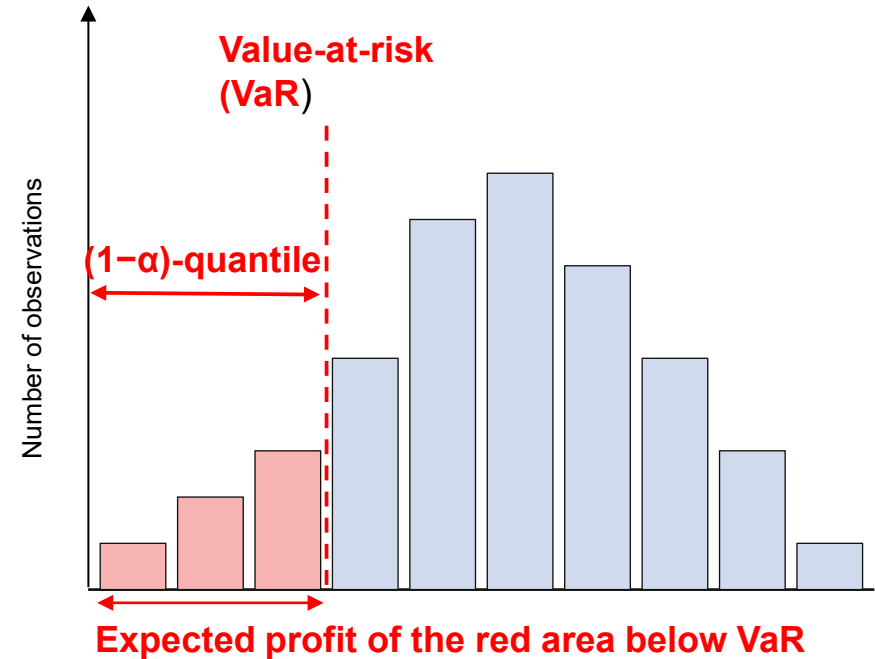
Conditional value-at-risk (CVaR): average of the bad tail



Reformulation of the CVaR-maximisation problem:

$$\begin{aligned}
 & \underset{P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}, \zeta, \eta_{\omega}}{\text{maximize}} && \zeta - \frac{1}{1-\alpha} \sum_{\omega} \pi_{\omega} \eta_{\omega} \quad \text{An auxiliary variable (€)} \\
 & \text{subject to} && f(P^{\text{DA}}) \leq 0, \\
 & && g(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \leq 0, \quad \forall \omega, \\
 & && \eta_{\omega} \geq 0, \quad \forall \omega, \\
 & && \eta_{\omega} \geq \zeta - \text{Profit}_{\omega}(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}), \quad \forall \omega.
 \end{aligned}$$

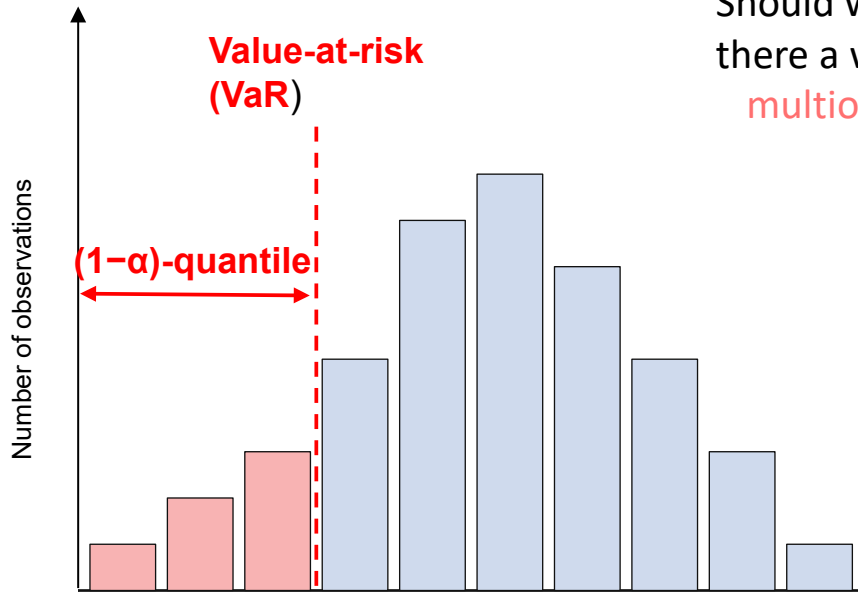
Conditional value-at-risk (CVaR): average of the bad tail



Reformulation of the CVaR-maximisation problem:

$$\begin{aligned}
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 & \text{subject to} && f(P^{\text{DA}}) \leq 0, \\
 & && g(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \leq 0, \quad \forall \omega, \\
 & && \eta_{\omega} \geq 0, \quad \forall \omega, \\
 & && \eta_{\omega} \geq \zeta - \text{Profit}_{\omega}(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}), \quad \forall \omega.
 \end{aligned}$$

Expected profit vs. CVaR



Should we maximize expected profit for CVaR? Is there a way to consider both? **Yes, let's formulate a multiobjective optimization problem!**

Revised objective function:

Given non-negative weight (parameter)

$$\underset{P^{DA}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}, \zeta, \eta_{\omega}}{\text{maximize}} \quad \underbrace{\left[\sum_{\omega} \pi_{\omega} \text{Profit}_{\omega}(P^{DA}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \right]}_{\text{Expected profit (€)}} + \beta \underbrace{\left[\zeta - \frac{1}{1-\alpha} \sum_{\omega} \pi_{\omega} \eta_{\omega} \right]}_{\text{CVaR (€)}}.$$

Expected profit (€)

CVaR (€)

Two possible approaches for weighting in the objective function

1) For $\beta \geq 0$

$$\underset{P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}, \zeta, \eta_{\omega}}{\text{maximize}} \quad \left[\sum_{\omega} \pi_{\omega} \text{Profit}_{\omega}(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \right] + \beta \left[\zeta - \frac{1}{1-\alpha} \sum_{\omega} \pi_{\omega} \eta_{\omega} \right].$$

2) For $0 \leq \beta \leq 1$

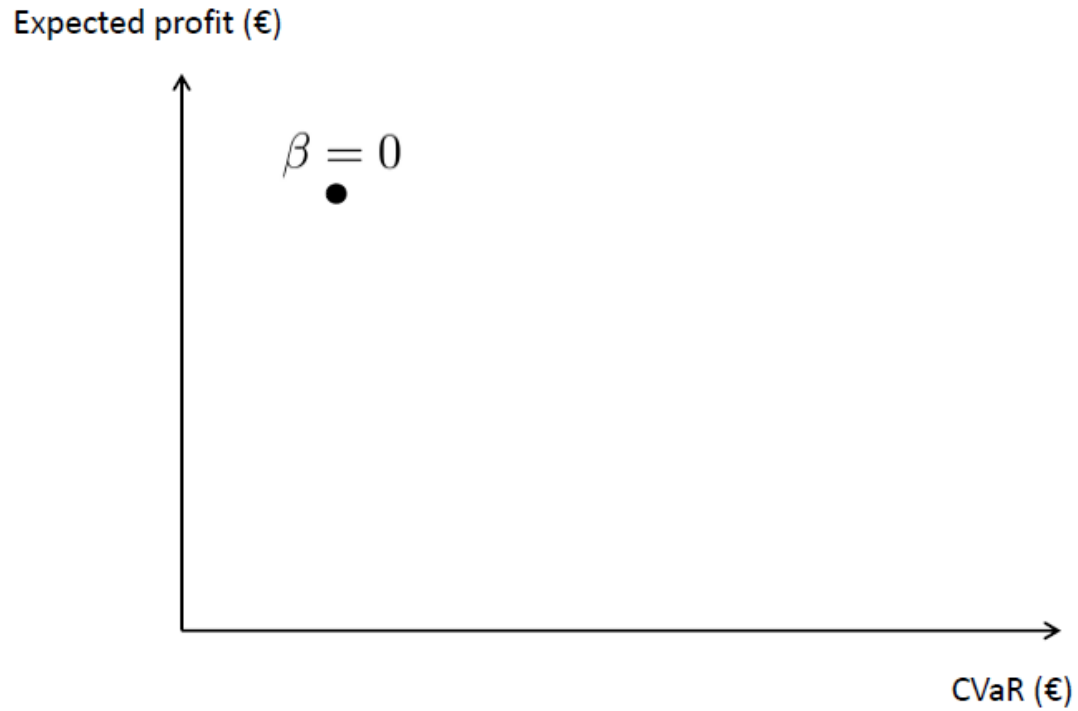
$$\underset{P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}, \zeta, \eta_{\omega}}{\text{maximize}} \quad (1 - \beta) \left[\sum_{\omega} \pi_{\omega} \text{Profit}_{\omega}(P^{\text{DA}}, \Delta_{\omega}^{\uparrow}, \Delta_{\omega}^{\downarrow}) \right] + \beta \left[\zeta - \frac{1}{1-\alpha} \sum_{\omega} \pi_{\omega} \eta_{\omega} \right].$$

Both versions of the weighted objective function can be used.

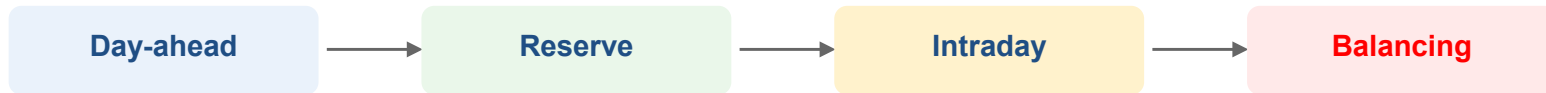
In both, $\beta = 0$ implies a **risk-neutral** case (expected profit maximization only).

In both, higher values of β imply greater conservativeness.

Efficient frontier



Extension: multi-market bidding portfolio



$$\text{portfolio_profit}[\omega] = \text{revenue_DA}[\omega] + \text{revenue_reserve}[\omega] + \text{revenue_intraday}[\omega] + \text{balancing_settlement}[\omega] - \text{penalties}[\omega]$$

Apply CVaR to $\text{portfolio_profit}[\omega]$, not to each market separately.

Coupled decisions

Capacity sold in one market reduces availability in another.

Obligation risk

Reserve and DA commitments create delivery obligations.

Portfolio view

CVaR rewards hedging between markets and technologies.

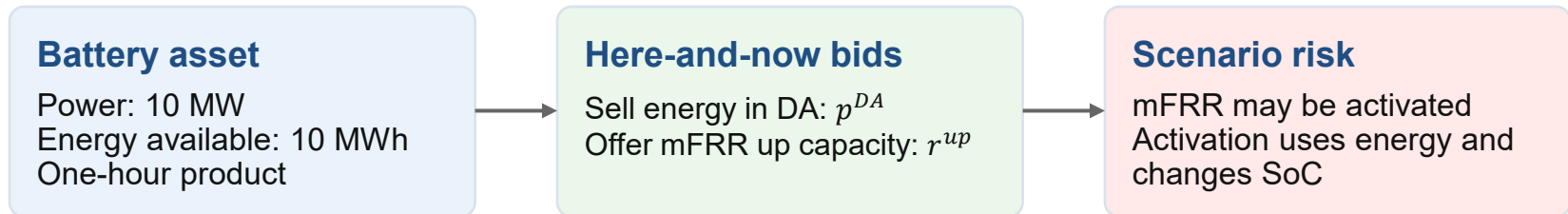
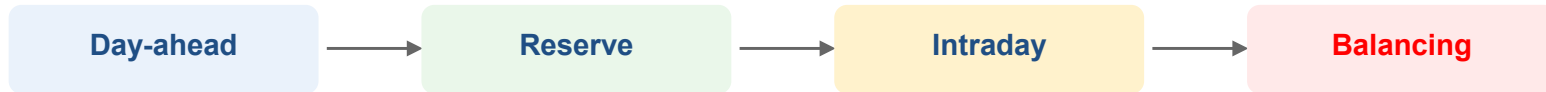
Risk-averse price-taking offering model

New additions (all other parts remain unchanged):

$$\begin{aligned}
 & \max_{p_t^{\text{DA}}, \delta_{t,\omega}^+, \delta_{t,\omega}^-, \zeta, \eta_\omega} \quad (1 - \beta) \sum_{\omega \in \Omega} \pi_\omega \text{Profit}_\omega + \beta \left(\zeta - \frac{1}{1 - \alpha} \sum_{\omega \in \Omega} \pi_\omega \eta_\omega \right) \\
 & \text{s.t.} \quad 0 \leq p_t^{\text{DA}} \leq P^{\text{max}}, \quad \forall t \in \mathcal{T}, \\
 & \quad \delta_{t,\omega} = p_{t,\omega}^{\text{real}} - p_t^{\text{DA}}, \quad \forall t \in \mathcal{T}, \omega \in \Omega, \\
 & \quad \delta_{t,\omega} = \delta_{t,\omega}^+ - \delta_{t,\omega}^-, \quad \forall t \in \mathcal{T}, \omega \in \Omega, \\
 & \quad \delta_{t,\omega}^+ \geq 0, \quad \delta_{t,\omega}^- \geq 0, \quad \forall t \in \mathcal{T}, \omega \in \Omega, \\
 & \quad \text{Profit}_\omega = \sum_{t \in \mathcal{T}} (\lambda_{t,\omega}^{\text{DA}} p_t^{\text{DA}} + \lambda_{t,\omega}^{\text{up}} \delta_{t,\omega}^+ - \lambda_{t,\omega}^{\text{down}} \delta_{t,\omega}^-), \quad \forall \omega \in \Omega, \\
 & \quad \eta_\omega \geq \zeta - \text{Profit}_\omega, \quad \forall \omega \in \Omega, \\
 & \quad \eta_\omega \geq 0, \quad \forall \omega \in \Omega.
 \end{aligned}$$

Here, π_ω denotes the probability of scenario ω , while $\beta \in [0, 1]$ determines the trade-off between expected profit and CVaR-based tail protection.

Battery example: bidding in DA + mFRR



Battery asset

Power: 10 MW
Energy available: 10 MWh
One-hour product

Here-and-now bids

Sell energy in DA: p^{DA}
Offer mFRR up capacity: r^{up}

Scenario risk

mFRR may be activated
Activation uses energy and changes SoC

Feasibility for upward mFRR:

$$p^{DA} + r^{up} \leq p^{\max}$$

$$p^{DA} + a_{\omega} r^{up} \leq E^{\text{available}} \quad \forall \omega \in \Omega$$

where a_{ω} is the activation fraction.

Scenario profit:

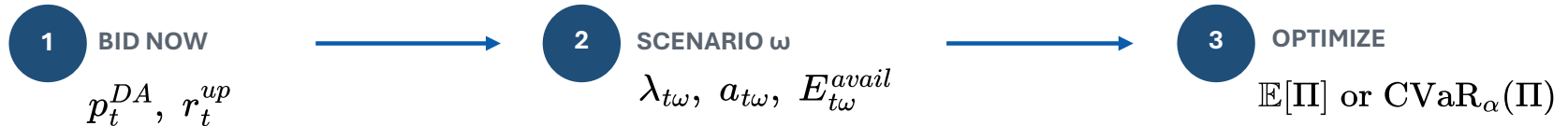
$$\begin{aligned} \Pi_{\omega} &= \lambda^{DA} p^{DA} + \lambda^{\text{cap}} r^{up} + \lambda_{\omega}^{\text{act}} a_{\omega} r^{up} \\ &\quad - \gamma_{\omega}, \quad \forall \omega \in \Omega \end{aligned}$$

Where γ_{ω} represents the penalty of not satisfying allocated capacity in scenario ω .

The same first-stage bid must remain feasible and profitable under uncertain mFRR activation.

Battery DA + mFRR: two-stage stochastic formulation

First-stage bids are fixed before uncertainty; scenario prices, activation and availability determine realized profit and tail risk.



1 First-stage bids and physical coupling

Decision variables

p_t^{DA} day-ahead discharge bid
 r_t^{up} upward mFRR capacity bid

Scenario-dependent states

$a_{t,\omega}$ activation fraction
 $e_{t,\omega}$ state of charge
 $\Pi_{t,\omega}$ realized portfolio profit

Physical limits

$$p_t^{DA} + r_t^{up} \leq P^{\max}$$

$$p_t^{DA} + a_{t,\omega} r_t^{up} \leq E_{t,\omega}^{avail}$$

2 Scenario profit and risk objective

Scenario profit

$$\Pi_\omega = \sum_t [\lambda_{t,\omega}^{DA} p_t^{DA} + \lambda_{t,\omega}^{cap} r_t^{up} + \lambda_{t,\omega}^{act} a_{t,\omega} r_t^{up} - \gamma_{t,\omega}]$$

Risk-neutral objective

$$\max \sum_{\omega \in \Omega} \pi_\omega \Pi_\omega$$

Risk-averse objective

$$\max (1 - \beta) \mathbb{E}[\Pi] + \beta \text{CVaR}_\alpha(\Pi)$$

mFRR capacity is attractive only when activation profit and energy feasibility remain robust across scenarios.

Battery example: energy-only vs DA + mFRR

Assume $\lambda^{\text{DA}} = 85 \text{ €/MWh}$, $\lambda^{\text{cap}} = 50 \text{ €/MW}$, activation price = 120 €/MWh, one-hour product.

Bid	p^{DA}	r^{up}	No activation	50% activation	100% activation	Expected profit	Worst outcome
A: DA only	10	0	850 €	850 €	850 €	850 €	850 €
B: DA + mFRR	5	5	675 €	975 €	1275 €	975 €	675 €

For three equiprobable scenarios, $\text{CVaR}_{2/3}$ equals the average of the worst one-third outcomes.

Risk-neutral view

Bid B has the highest expected profit because it earns capacity and activation revenue.

Risk-averse view

Bid A has a better worst outcome and higher $\text{CVaR}_{2/3}$ for the lower tail.

Key constraint

Avoid $p^{\text{DA}} + r^{\text{up}} > 10$ MW/MWh in full activation.

Multi-market bidding is not only revenue stacking; it also creates coupled delivery obligations and tail risk.

Battery example: expected profit vs tail protection

Bid	Mean	Worst	CVaR	Interpretation
A: DA only	850 €	850 €	850 €	stable but lower mean
B: DA + mFRR	975 €	675 €	675 €	higher mean, weaker tail

$\beta = 0$

The model selects the highest expected profit: likely Bid B.

β high

The model values downside protection: Bid A may dominate.

Operational insight

mFRR participation is attractive only if tail exposure is acceptable.

Problem description: battery bidding in DA + mFRR markets

Asset & markets

- Grid-connected battery (BESS): power limits, SOC limits, charge/discharge efficiencies, degradation cost.
- Participates in the day-ahead (DA) energy market and the mFRR market (capacity bids + energy activation).
- **When activated, the battery delivers up/down power toward the TSO's activation target.**
- 24 h horizon: 15-min energy steps ($T = 96$), hourly reserve bids ($H = 24$).

Decisions in two stages

Stage 1 (here-and-now, $D-1$):

- DA charge/discharge schedule.
- mFRR up/down capacity bids with min-bid rules and headroom/footroom limits.

Stage 2 (recourse, per scenario ω):

- Actual charging/discharging and SOC trajectory.
- Delivered activation vs. the TSO activation target; over-/under-delivery penalized.

Uncertainty & objective

Unknown at bid time:

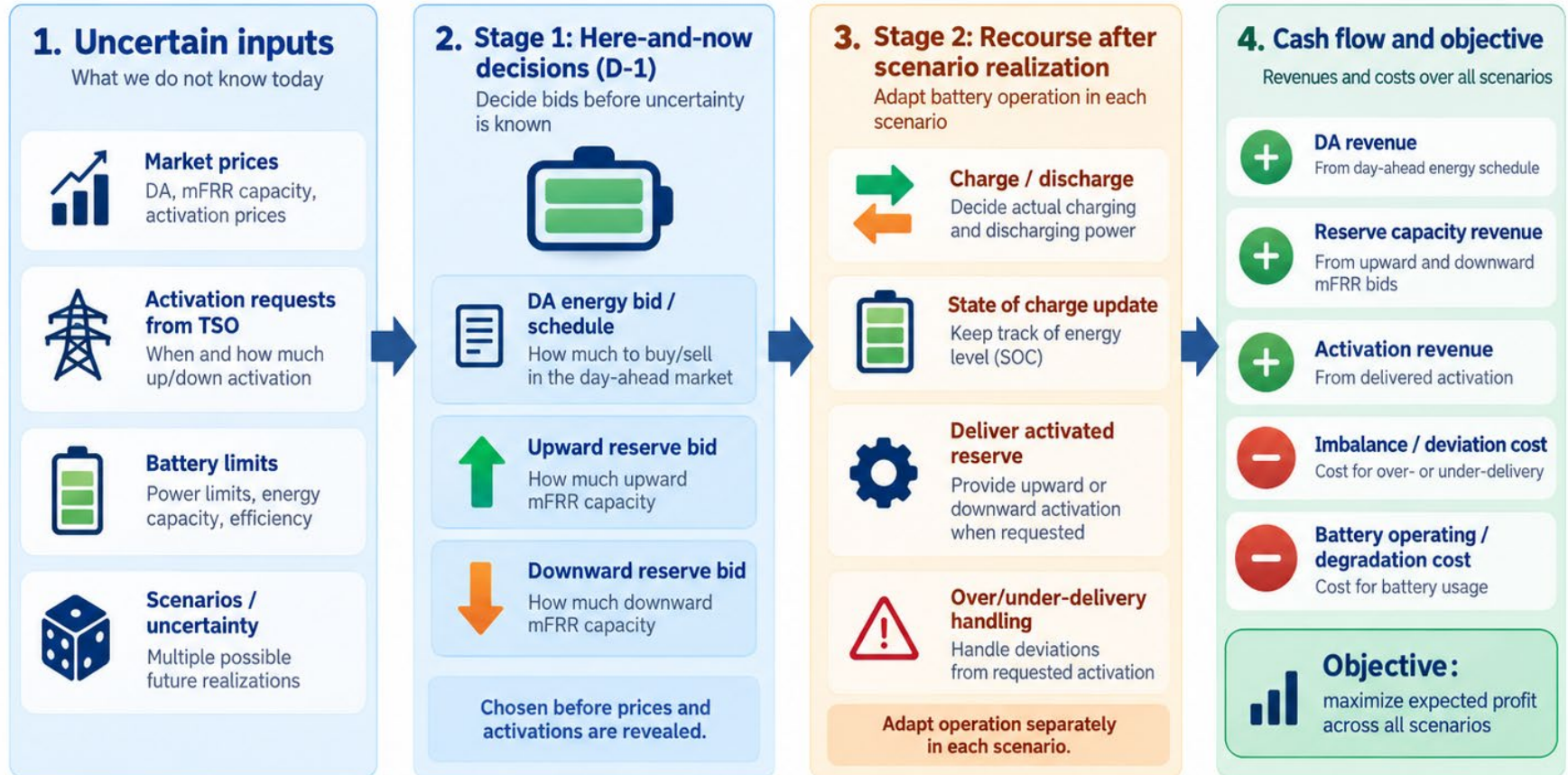
- DA prices, mFRR capacity & activation prices, TSO activation volumes.
- Modeled by scenarios $\omega \in \Omega$ with probabilities $\pi\omega$.

Goal:

- Maximize expected profit: DA + capacity + activation revenue – imbalance, tariff & degradation costs.
- Risk-averse variant: mean-CVaR trade-off with weight β .

Core difficulty: bids are fixed before prices and activations are realized → two-stage stochastic MILP (binaries for bid flags and charge/discharge exclusivity).

Overall idea: BESS bidding in DA+mFRR markets



$$\text{Expected profit} = \text{Revenues from DA} + \text{Revenues from reserves} + \text{Revenues from activation} - \text{Imbalance costs} - \text{Battery operating costs}$$

Key message: | The model commits bids first and then adjusts battery operation when uncertainty becomes reality.

Two-stage stochastic MILP: battery in DA + mFRR markets

Sets & parameters

Sets:

$T = \{0, \dots, T-1\}$: time steps $T=96$, $\Delta t=0.25$ h

$\mathcal{H} = \{0, \dots, H-1\}$: hours $H=24$

ω : scenarios, $\Pi_\omega = 1/|\Omega|$

Battery physics:

η_{ch}, η_{dis} : efficiencies
 $P_{ch}^{max}, P_{dis}^{max}$: power limits (MW)
 E_{min}, E_{max} : SOC limits
 E_{init} : initial/terminal SOC
 B_{min} : min bid, SoC margin

Exogenous (ω -indexed):

$\lambda_{t,\omega}^{DA}$: day-ahead price
 $\lambda_{t,\omega}^{up}, \lambda_{t,\omega}^{dn}$: mFRR prices
 $\lambda_{t,\omega}^{c,up}, \lambda_{t,\omega}^{c,dn}$: cap. prices
 $V_{t,\omega}^{up}, V_{t,\omega}^{dn}$: TSO target activation (MW)

Stage 1 (here-and-now, D-1)

Variables (not scenario-indexed):

$p_t^{DA} \in [-P_{ch}^{max}, P_{dis}^{max}]$, $\forall t \in T$: DA schedule
 $r_h^{up}, r_h^{dn} \geq 0$, $\forall h \in \mathcal{H}$: reserve cap bids
 $y_h^{up}, y_h^{dn} \in \{0,1\}$, $\forall h \in \mathcal{H}$: bid flags

Constraints:

$$\begin{aligned} -P_{ch}^{max} &\leq p_t^{DA} \leq P_{dis}^{max} \\ B_{min} y_h^{up} &\leq r_h^{up} \leq P_{dis}^{max} y_h^{up} \\ B_{min} y_h^{dn} &\leq r_h^{dn} \leq P_{ch}^{max} y_h^{dn} \end{aligned}$$

Headroom:

$$p_t^{DA} + r_h^{up} \leq P_{dis}^{max}$$

Footroom:

$$p_t^{DA} - r_h^{dn} \geq -P_{ch}^{max}$$

No simultaneous reserve direction:

$$y_h^{up} + y_h^{dn} \leq 1, \quad \forall h \in \mathcal{H}$$

Decision timing: DA schedule and mFRR capacity bids are chosen before prices/activations are realized.

Stage 2 (recourse, per scenario ω)

Variables (scenario-indexed):

$p_{t,\omega}^{ch}, p_{t,\omega}^{dis} \geq 0$: Charging and discharging power
 $e_{t,\omega} \in [E_{min}, E_{max}]$: SOC
 $d_{t,\omega}^+, d_{t,\omega}^- \geq 0$: Over/Under-delivery
 $u_{t,\omega} \in \{0,1\}$: No-sim. Flag

SOC dynamics $\forall t, \forall \omega$:

$$e_{t+1,\omega} = e_{t,\omega} + \eta_{ch} \Delta t p_{t,\omega}^{ch} - \frac{\Delta t}{\eta_{dis}} p_{t,\omega}^{dis}$$

$$e_{0,\omega} = e_{T,\omega} = e^{init}, \quad \forall \omega \in \Omega.$$

Power link $\forall t, \forall \omega$:

$$p_{t,\omega}^{dis} - p_{t,\omega}^{ch} = p_t^{DA} + V_{t,\omega}^{up} - V_{t,\omega}^{dn} + d_{t,\omega}^+ - d_{t,\omega}^-,$$

$$0 \leq d_{t,\omega}^+ \leq D^{max}: \text{ over-delivery relative to the target,}$$

$$0 \leq d_{t,\omega}^- \leq D^{max}: \text{ under-delivery relative to the target.}$$

$$D^{max} = P_{ch}^{max} + P_{dis}^{max} \quad \text{Maximum deviation bound (MW)}$$

No simultaneous charge/discharge:

$$p_{t,\omega}^{ch} \leq P_{max}^{ch} (1 - u_{t,\omega}),$$

$$p_{t,\omega}^{dis} \leq P_{max}^{dis} u_{t,\omega}.$$

SoC adequacy for reserve bids:

$$e_{t,\omega} - E_{min} \geq \frac{\tau}{\eta_{dis}} r_{h(t)}^{up}, \quad [\text{energy available for discharge}],$$

$$E_{max} - e_{t,\omega} \geq \eta_{ch} \tau r_{h(t)}^{dn}, \quad [\text{capacity available to absorb energy}].$$

$V_{t,\omega}^{up}, V_{t,\omega}^{dn}$ are exogenous activation trajectories, not decision variables.

Objective (risk-neutral):

$$\max \sum_{\omega \in \Omega} \pi_\omega \left\{ \Delta t \sum_t \lambda_{t,\omega}^{DA} p_t^{DA} + \sum_h (\lambda_{h,\omega}^{c,up} r_h^{up} + \lambda_{h,\omega}^{c,dn} r_h^{dn}) + \right. \\ \left. \Delta t \sum_t (\lambda_{t,\omega}^{up} V_{t,\omega}^{up} - \lambda_{t,\omega}^{dn} V_{t,\omega}^{dn}) + \Delta t \sum_t (\lambda_{t,\omega}^{up} d_{t,\omega}^+ - \lambda_{t,\omega}^{dn} d_{t,\omega}^-) - \Delta t \sum_t [c^{tariff} p_{t,\omega}^{ch} + c^{deg} (p_{t,\omega}^{ch} + p_{t,\omega}^{dis})] \right\}$$

CVaR extension: mean–CVaR objective for risk-averse bidding

Scenario profit (new variable)

$$\text{Profit}(\omega) = \text{DA_rev}(\omega) + \text{Cap_rev}(\omega) + \text{Act_rev}(\omega) + \text{Dev_rev}(\omega) - \text{Cost}(\omega)$$

$$\text{DA_rev}(\omega) = \Delta \sum_t \pi_{t,\omega}^{\text{DA}} p_t^{\text{DA}}$$

$$\text{Cap_rev}(\omega) = \sum_h (\pi_{h,\omega}^{\text{c,up}} r_h^{\text{up}} + \pi_{h,\omega}^{\text{c,dn}} r_h^{\text{dn}})$$

$$\text{Act_rev}(\omega) \& = \Delta t \sum_t (\lambda_{t,\omega}^{\text{up}} V_{t,\omega}^{\text{up}} - \lambda_{t,\omega}^{\text{dn}} V_{t,\omega}^{\text{dn}}) + \Delta t \sum_t (\lambda_{t,\omega}^{\text{up}} d_{t,\omega}^+ - \lambda_{t,\omega}^{\text{dn}} d_{t,\omega}^-)$$

$$\text{Cost}(\omega) \& = \Delta t \sum_t [c^{\text{tariff}} p_{t,\omega}^{\text{ch}} + c^{\text{deg}} (p_{t,\omega}^{\text{ch}} + p_{t,\omega}^{\text{dis}})]$$

CVaR reformulation (Rockafellar & Uryasev, 2000)

Auxiliary variables:

$\zeta \in \mathbb{R}$ (free variable) $\leftarrow$ VaR level
 $\xi_\omega \geq 0, \quad \forall \omega \in \Omega$ $\leftarrow$ shortfall below VaR

Shortfall constraint $\forall \omega \in \Omega$:

$$\xi_\omega \geq \zeta - \text{Profit}_\omega,$$

$$\xi_\omega \geq 0, \quad \forall \omega \in \Omega.$$

CVaR $_\alpha$ expression:

$$\text{CVaR}_\alpha = \zeta - \frac{1}{1-\alpha} \sum_{\omega \in \Omega} \pi_\omega \xi_\omega$$

Mean–CVaR objective

$$\max \quad (1-\beta) \sum_{\omega \in \Omega} \pi_\omega \text{Profit}_\omega + \beta \left[\zeta - \frac{1}{1-\alpha} \sum_{\omega \in \Omega} \pi_\omega \xi_\omega \right]$$

Subject to:

- All stage-1 and stage-2 constraints
- Profit definition (equality constraints)
- CVaR shortfall constraints

Risk parameters:

$\alpha \in (0,1)$ confidence level (typical values: 0.90, 0.95, 0.99)

CVaR $_\alpha$ = avg profit of worst $(1-\alpha)$ fraction

$\beta \in [0,1]$ risk-aversion weight

$\beta = 0 \rightarrow$ risk-neutral: $\max E[\text{Profit}]$

$\beta = 1 \rightarrow$ fully risk-averse: $\max \text{CVaR}_\alpha$

$0 < \beta < 1 \rightarrow$ mean–CVaR trade-off

Additions to risk-neutral model:

- +1 free variable (ζ)
- + $|\Omega|$ variables ($\xi_\omega \geq 0$)
- + $|\Omega|$ constraints (shortfall)
- + $|\Omega|$ constraints (Profit definition)

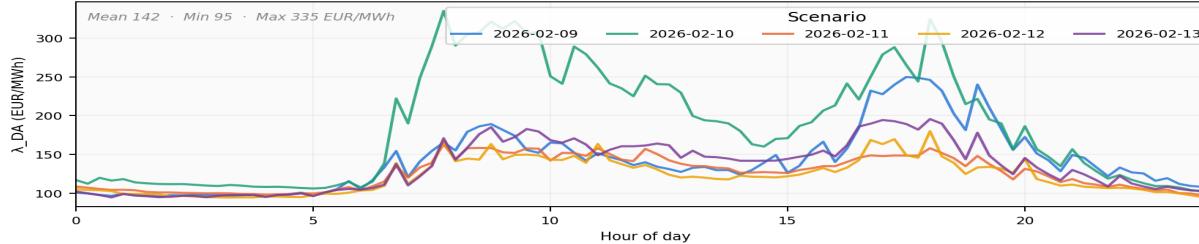
same MILP solver applies.

$\beta > 0$: same stage-1/stage-2 constraints + two new variable blocks + one new objective term.

NO3 Input data overview

NO3 area, 5 scenarios (09–13 Feb 2026) · DA · mFRR activation · reserve capacity prices · net activated volume

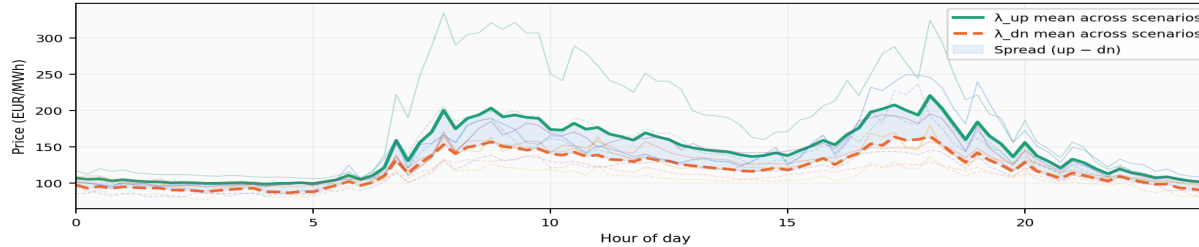
(A) Day-ahead prices — 15-min resolution



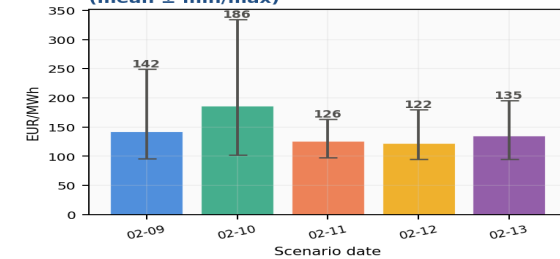
(B) Dataset summary

Scenarios	5 days (Feb 2026)
Resolution DA	15-min (T = 96 steps)
Res. cap.	hourly (H = 24 steps)
Time step Δt	0.25 h
DA price range	95 – 335 EUR/MWh
DA price mean	142.5 EUR/MWh
mFRR up range	95 – 335 EUR/MWh
mFRR dn range	78 – 237 EUR/MWh
Cap up range	1.00 – 25.70 EUR/MW
Cap dn range	1.93 – 4.10 EUR/MW
NAV range	-683 – 111 MW
NAV mean	-136.4 MW (net downward)

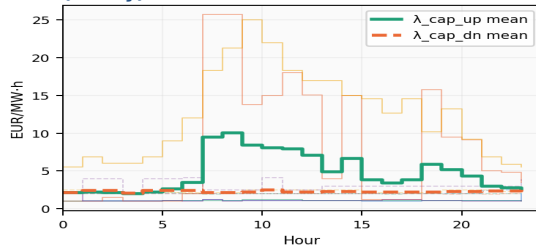
(C) mFRR activation prices — 15-min resolution



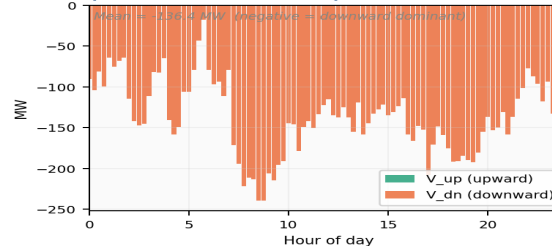
(D) DA price per scenario (mean ± min/max)



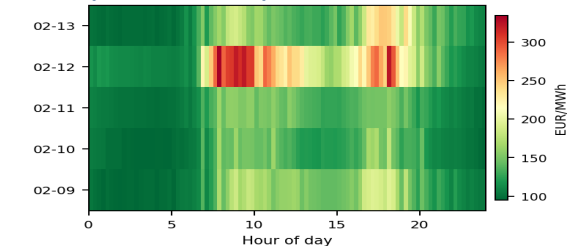
(E) Capacity prices (hourly)



(F) Net activated volume (mean across scenarios)



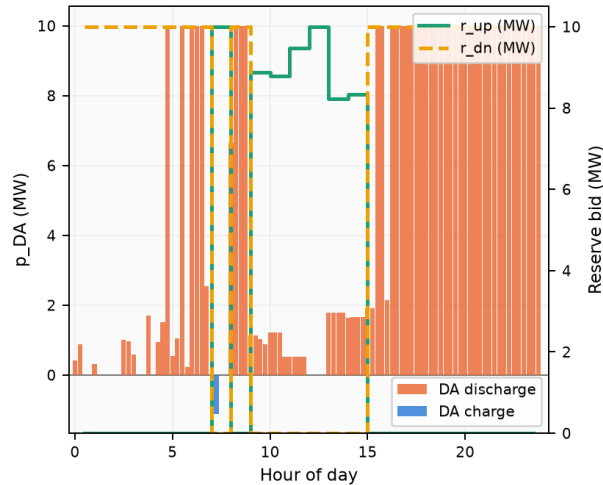
(G) DA price heatmap (scenario × hour)



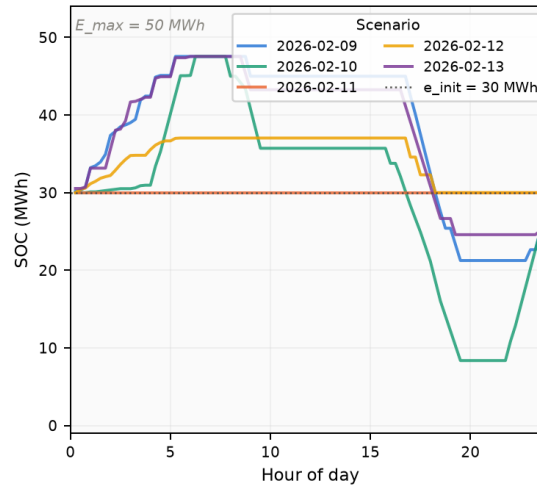
Battery in DA+mFRR multi Market

10 MW/50 MWh, 5 Scenarios, $\alpha = 0.95$ $\beta = 0.5$

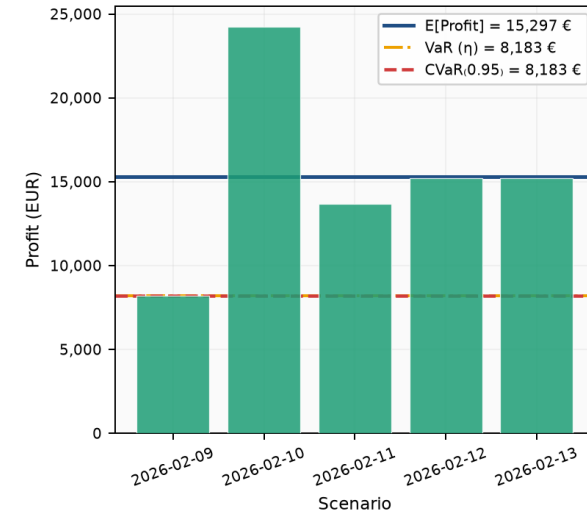
(A) Stage-1 bids



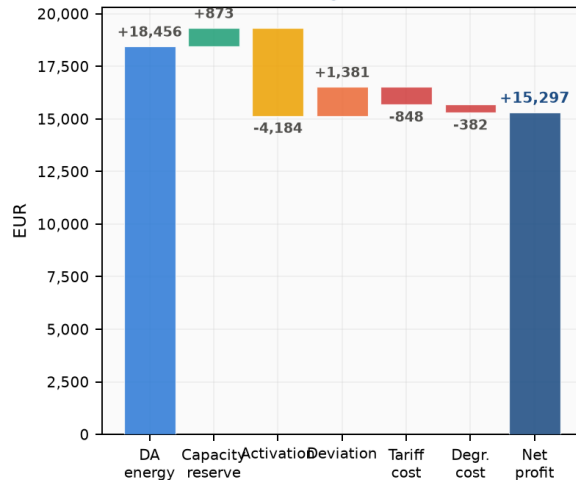
(B) Battery SOC trajectories



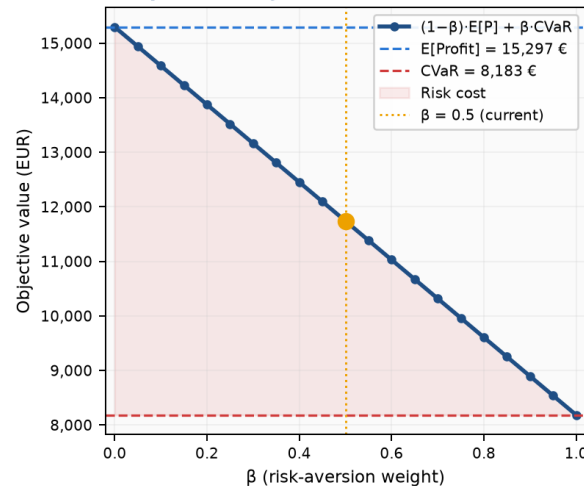
(C) Scenario profit distribution



(D) Revenue stacking (mean, EUR)



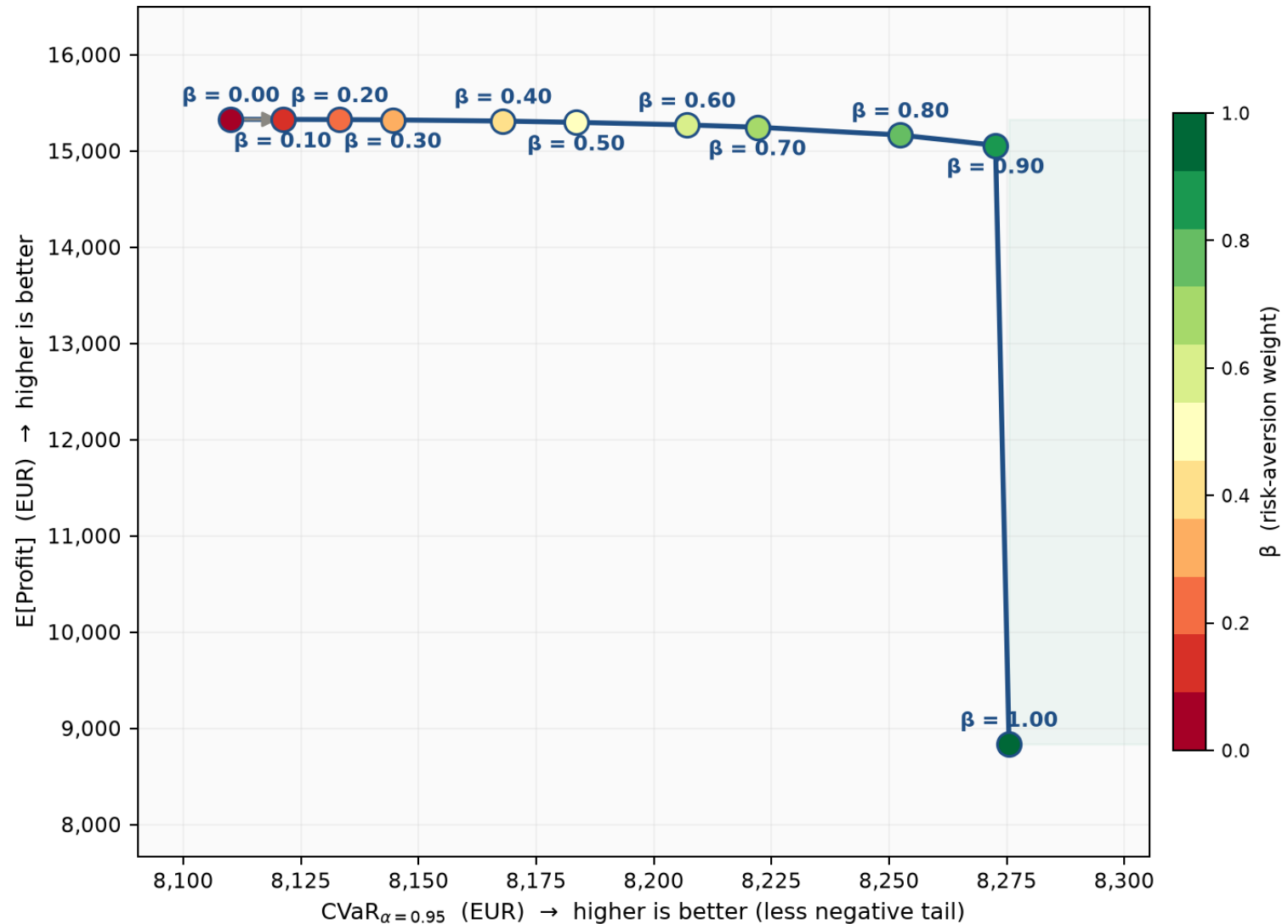
(E) β sensitivity (mean-CVaR trade-off)



(F) Model summary & risk measures

Battery	10 MW / 50 MWh
Scenarios	5 ($\pi_w = 0.200$)
α (confidence)	0.95 $\rightarrow$ worst 5%
β (risk weight)	0.50 $\rightarrow$ risk-averse
E[Profit]	+15,297.0 EUR
VaR(η)	+8,183.5 EUR
CVaR(0.95)	+8,183.5 EUR
Objective	+11,740.2 EUR
Best scenario	+24,237.7 EUR
Worst scenario	+8,183.5 EUR
Max r_{up} bid	10.00 MW ($h=7$)
Max r_{dn} bid	10.00 MW ($h=0$)

Efficient frontier (expected profit vs CVaR)



What changes when the bidder becomes risk-averse?

Risk-neutral

May offer close to expected production if expected settlement is favorable.

higher expected profit

Risk-averse

May withhold volume to avoid expensive shortage in low-wind scenarios.

higher tail profit

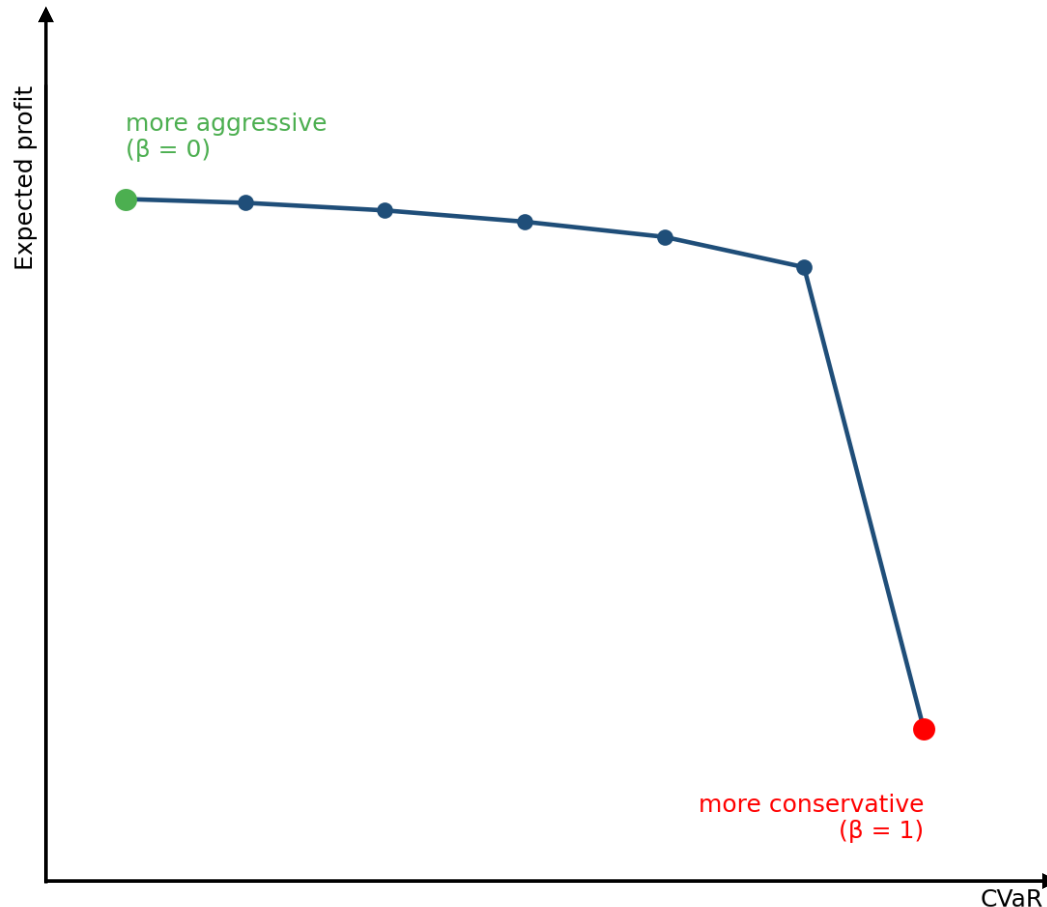
Multi-market

May allocate capacity to reserves or intraday as a hedge.

diversified revenue

Risk aversion does not mean “bid less” in every hour; it means “bid differently where the tail is dangerous”.

Trade-off: expected profit vs. CVaR



β sweep

Solve the model for $\beta = 0, 0.25, 0.5, 0.75, 1$.

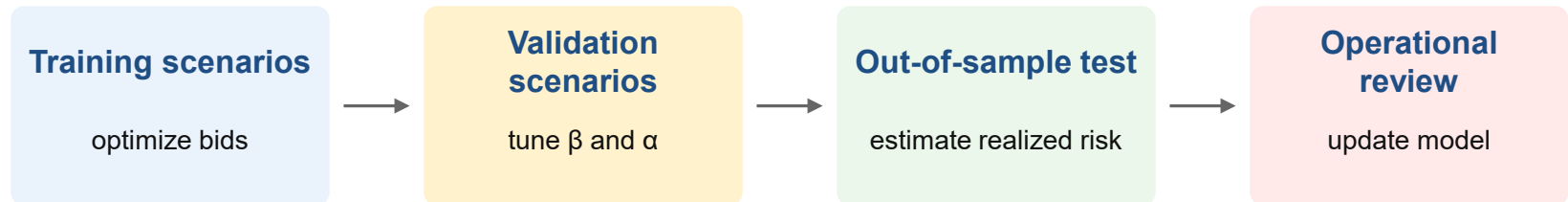
Interpretation

The efficient frontier shows the cost of risk reduction.

Decision

Pick β based on risk appetite or mandated risk limit.

Testing the quality of risk-aware decisions



In-sample danger

A decision can look excellent on the scenarios used for optimization.

Out-of-sample discipline

Evaluate on unseen historical days or simulated scenarios.

Key takeaways

- Risk management is about the entire profit distribution, not only its average.
- VaR identifies the tail threshold; CVaR measures the average outcome in the bad tail.
- CVaR has a linear reformulation for scenario-based stochastic programming.
- A mean-CVaR objective provides a transparent trade-off between profit and downside protection.
- In multi-market bidding, risk should be evaluated on portfolio-level cash flow.